

ELLIOTT WAVE / NEOWAVE

Developer Coding Manual & Required Pine Script Updates

Prepared for a Pine Script / Python developer who does not need prior Elliott Wave knowledge.

Version: Developer Handoff - August 2026

Purpose

This manual explains the Elliott concepts, exact coding states, formulas, invalidation logic, correction classifiers, Fibonacci/time/MACD rules, controlled recount behavior and acceptance tests required to upgrade the current 1,007-line Elliott overlay into a state-based automatic Elliott Wave engine.

Important

Where the supplied source notes disagree with each other, this manual does NOT silently invent a rule. Those items are marked OWNER DECISION / DO NOT GUESS and must remain configurable or blocked until approved.

Executive Summary

The current Pine Script contains useful components: confirmed pivot detection, ATR/range swing filtering, Important High/Low context, MACD/RSI values at pivots, a 61.8% degree concept, several Fibonacci levels, partial Wave-2/Wave-3/Wave-4/Wave-5 checks and a time-rule table. These should be reused.

The central problem is that the script still assigns Elliott labels mainly by swing sequence / cycle position and then checks rules afterward. A true Elliott engine must do the reverse: build a candidate, validate price structure first, then Fibonacci/time/internal structure, then use MACD/RSI/trendlines as supporting evidence, and only then confirm the wave label.

One sentence for the developer

Do not ask "what label comes next in the sequence?" Ask "which Elliott candidate(s) can this new confirmed pivot legally complete?"

Required Processing Order

```

RAW PRICE / CONFIRMED PIVOTS
  |
  v
IMPORTANT / DEGREE FILTER
  |
  v
BASE CANDIDATE -> BASE CONFIRMED / LOCKED
  |
  v
STRUCTURAL WAVE / CORRECTION CANDIDATES
  |
  v
HARD PRICE RULES
  |
  v
INTERNAL STRUCTURE
  |
  v
FIBONACCI + TIME
  |
  v
MACD / RSI / TRENDLINE / GAP SUPPORT
  |
  v
FORMING / CONFIRMED / ALTERNATE / INVALID
  |
  v
FINAL LABEL

```

Priority Summary

Priority	Required change	Reason
P0	Replace fixed/modulo wave cycle with candidate state machine	Without this, later rules only create warnings; they do not control the count.
P0	Separate raw pivots from Elliott pivots	Small internal swings must not advance the main-degree count.
P0	Lock Point 0 / Base after confirmation	Prevents anchor drift and uncontrolled recount.
P0	Make hard invalidation reject/recount	An invalid wave must not remain as a red but accepted label.

Priority	Required change	Reason
P0	Fix 144-day/degree Important H/L handling	144 chart bars is not 144 days on intraday charts.
P1	Implement Wave 1-5 structural engines	Core impulse logic.
P1	Automatic correction classifier	Zig-Zag, Flat, Double, Triple and Triangle cannot be manual label choices.
P1	Leading / Ending Diagonal + extensions	Needed for legitimate overlap and special Wave-5/C behavior.
P2	Wave-specific Fib/time/MACD/trendlines	Required for confirmation and debugging.
P2	Controlled recount / non-repaint / alternate counts	Required for live trading stability.
P3	Gaps, candlestick support, production debug report	Final supporting modules and validation.

How to Read This Manual

- SOURCE RULE = rule taken from the supplied Elliott/NeoWave notes/specification.
- PROJECT REQUIREMENT = additional behavior required for this project, such as HP BUY/SELL and FBD/FBO controlled recount.
- HARD RULE = violation invalidates that candidate.
- GUIDELINE / SUPPORT = affects rank or confidence, but should not override valid price structure unless the approved source explicitly makes it mandatory.
- OWNER DECISION = the supplied source contains conflicting or incomplete instructions; do not guess.

Source Hierarchy and Conflict Policy

The coding rules in this manual are grounded primarily in the supplied "EW Notes / Chartking" material and the derived "Elliott Wave Indicator - Developer Requirement Specification". Some additional time-analysis material contains alternative values. When two source sections disagree, preserve the conflict instead of silently reconciling it.

OWNER DECISION / DO NOT GUESS

Examples of unresolved source conflicts include Wave-2 time, some Wave-4 retracement limits during extensions, precise terminal/trending Wave-3 boundaries, Wave-C elongated maximum, Double Correction confirmation, selected X/XX/Z rules, LD/ED coexistence wording, exact MACD parameters and the exact pivot definition. Implement a visible conflict mode or keep the affected candidate FORMING/TBD-BLOCKED until the owner chooses the rule.

PART I - Elliott Wave Concepts for a Programmer

1. Core Terminology

Term	Meaning for code
Point 0 / Base	Starting pivot of the current degree impulse/correction. It must be selected and then locked.
Impulse	Directional sequence normally described as Wave 1-2-3-4-5.
Correction	Counter-trend structure after/inside impulse: ABC, WXY, WXY-XX-Z, triangle, etc.

Term	Meaning for code
Degree	Scale of the wave. Main-degree pivots must be separated from subordinate/internal pivots.
Internal structure	The smaller waves that compose a larger wave. A large wave cannot be confirmed only from one start and one end pivot.
Candidate	A possible Elliott interpretation that is not yet confirmed.
Confirmed pivot	A pivot that has passed the required confirmation rules and should normally remain locked.
Alternate count	Secondary still-valid Elliott interpretation retained while the market is ambiguous.
Hard invalidation	A price/structure event that makes a specific candidate impossible.
Supporting evidence	MACD/RSI divergence, gaps, candlesticks, trendline/channel behavior, etc., used after structure.

2. State Model

State	Meaning	Drawing behavior
SEARCHING	No valid base/current candidate	No Elliott label or only debug marker.
FORMING	Enough structure exists for a possible wave, but required confirmation is incomplete	Optional DEV/FORMING label; may move/recount.
CONFIRMED	Required structural/Fib/time/confirmation conditions met	Lock main label unless later hard structural invalidation permits controlled recount.
ALTERNATE	Secondary valid interpretation remains possible	Show only in debug/alternate mode.
INVALID	Hard source rule broken	Reject candidate; store reason; do not keep as accepted red wave label.
TBD-BLOCKED	Source itself is blank/conflicting	Do not invent a rule; keep visible in debug.

3. Required Data Model

```
WaveCandidate {
    degree, timeframe, direction, label, patternType, subtype,
    startBar, endBar, startTime, endTime, startPrice, endPrice,
    durationBars, priceLength, fibRetrace, fibExtension,
    internalCount, internalPattern,
    macdLine, macdHist, rsi, divergenceState,
    trendlineState, channelState, gapState, candleState,
    status, hardInvalid, invalidReason, sourceRuleId,
    parentWaveId, childWaveIds[]
}
```

Also maintain raw pivot records separately from Elliott candidates. The same raw pivot may be used as an internal sub-wave without becoming a main-degree Elliott pivot.

PART II - Pivot, Degree and Base Engine

4. Raw Pivot Engine

The current pivot/ATR filtering infrastructure can be retained as a raw swing provider. However, raw pivot acceptance must not directly assign a main Elliott label.

Required rule

A raw pivot is market-structure data. An Elliott pivot is a validated semantic event. Keep these as different objects/arrays.

```
onNewConfirmedRawPivot(p):
    updateRawSwingBuffer(p)
    updateCurrentFormingLeg(p)
    evaluateExistingWaveCandidates(p)
    proposeNewCandidatesOnlyWhereContextAllows(p)
    neverAdvanceMainWaveNumberOnlyBecausePivotCountIncreased
```

5. Important High / Low and 144-Day Context

The current script uses a numeric lookback of 144 chart bars. On a 15-minute chart that is only 36 hours, so it cannot represent a 144-day source context. The Important H/L scanner must be degree/timeframe-aware.

```
importantHigh = highest(sourceDegreeHigh, lookbackEquivalentTo144Days)
importantLow  = lowest(sourceDegreeLow, lookbackEquivalentTo144Days)
```

Recommended implementation: calculate the Important H/L on the approved source/degree timeframe using request.security(), and store the source timeframe, timestamp and price with the base candidate.

6. Base / Point 0 Candidate

A bullish base is an important low from which a degree-up impulse may begin. A bearish base is the mirrored important high.

Source basis: Wave-1 source framework: Important H/L context, MACD lowest low or positive divergence, and minimum 61.8% degree movement before confirming the degree Wave-1.

```
BullishBaseCandidate = importantLowContext
                      AND directionCanMoveUp
                      AND degreePotential
                      AND oscillatorEvidenceConfigured

BearishBaseCandidate = mirrored logic
```

Do not make "MACD extreme" and "MACD/RSI divergence" sound identical. The current script allows MACD extreme OR divergence. Make this user-selectable and visible in debug output.

7. Base Lock and Controlled Release

```
BASE_CANDIDATE -> BASE_CONFIRMED -> BASE_LOCKED
```

Release BASE_LOCKED only when:

- hard Elliott invalidation occurs, OR
- higher-degree completion / degree fall requires a new origin, OR
- approved FBD/FBO structural recount candidate proves superior, OR
- existing count becomes structurally impossible.

PROJECT REQUIREMENT

FBD/FBO may nominate a new Elliott origin/recount candidate, but must never blindly reset the Elliott count. The existing confirmed count remains until the new origin passes degree/structure tests or the old count is invalidated.

PART III - Impulse Engine: Wave 1 to Wave 5

8. Wave 1 Coding Algorithm

Source basis: Wave 1: impulsive, five subordinate moves / 5-3-5-3-5 internal framework; MACD lowest low or positive divergence at the base; degree confirmation after minimum 61.8% Important H/L movement; Leading Diagonal may occur in Wave 1.

```
W1 candidate:
    P0 = locked base
    build subordinate impulse candidate from P0
    track degree progress = abs(currentPrice - P0) / importantRange

if degreeProgress < 0.618:
    status = FORMING
else if internalImpulseValid and hardRulesPass:
    status = CONFIRMED
else:
    remain FORMING / INVALID depending on failed rule
```

Do not do this

Do not call the first raw pivot after P0 "Wave 1" simply because it is the first swing.

9. Wave 2 Coding Algorithm

Source basis: Normal Wave-2 retracement: 23.6%-81.2% of W1. Special microscopic clue around 14% is treated separately and is associated with an extended W3 possibility. W2 may form ABC or WXY; MACD generally falls slowly and should not violate the approved base-MACD condition.

```
For bullish impulse:
    W1Length = P1 - P0
    W2Retrace = (P1 - currentLow) / W1Length

For bearish impulse:
    W1Length = P0 - P1
    W2Retrace = (currentHigh - P1) / W1Length
```

Retracement	Classification	Action
~14% special case	W2_MICROSCOPIC candidate	Do not use as normal minimum; prepare W3-extension context.
23.6%-61.8%	Normal W2 range	Keep W2 candidate if internal correction valid.
61.8%-81.2%	Deep W2 / project HP zone	Still requires correction completion and confirmation.
>81.2% but Point 0 not broken	Outside normal range	Warn / apply owner-selected source rule; do not automatically call normal W2.
Crosses Point 0	Hard invalid	Reject bullish/bearish impulse candidate and controlled recount.

10. Wave 2 Internal Structure

Wave 2 is a correction, not merely the next opposite pivot. Maintain correction candidates underneath W2: Zig-Zag, Flat and W-X-Y where context allows.

W2_FORMING

- | - ZigZag candidate (A-B-C)
- | - Flat candidate (A-B-C)
- ` - WXY candidate

Confirm W2 only when one correction candidate completes and the Point 0 invalidation is preserved.

11. HP BUY / HP SELL - Project Trading Signal

PROJECT REQUIREMENT

The 61.8%-81.2% Wave-2 zone is a candidate zone, not an automatic trade. HP BUY/SELL requires Wave-2 corrective completion plus confirmation. Base/Point 0 is the structural invalidation reference. The project may also use support/AVWAP/Ripple confirmation and a confirmation candle; these should be implemented as configurable project filters rather than changing the Elliott count itself.

HP BUY candidate:

- bullish W1 CONFIRMED
- AND W2 is valid correction
- AND W2 retracement in 61.8%-81.2% zone
- AND Point 0 remains valid
- AND correction completion / reversal confirmation
- AND project confirmation filters pass

HP SELL = exact bearish mirror

Chart should show W1, W2, 61.8%, 81.2%, HP zone, confirmation state and invalidation. Do not print HP BUY/SELL just because price touches a Fib line.

12. Wave 3 Coding Algorithm

Source basis: Wave 3 is impulsive; source main degree rule uses minimum 161.8% of the 0-1-2 extension, while terminal subordinate W3 may exist from the terminal minimum below 161.8%. W3 normally has the strongest/top MACD context and can extend to 9/13/17/21 moves.

```
W3Projection = extensionFrom(P0, P1, P2, currentPrice)

if W3Projection >= 1.618 and internalImpulseValid:
    subtypeCandidate = TRENDING_W3
elif terminalRulesPass and W3Projection >= terminalMin and W3Projection < 1.618:
    subtypeCandidate = TERMINAL_W3
else:
    do not confirm W3 yet
```

The current script treats below-161.8% mainly as a warning. Replace this with an explicit trending-versus-terminal classifier.

13. Wave 3 Time and MACD

Keep the existing W3 time relationships, but make the selected source version visible. One source version uses the average of W1/W2 when their times differ and the sum when they are approximately equal; other notes present a simpler minimum-time relationship. This is a conflict-sensitive area.

OWNER DECISION / DO NOT GUESS

Do not hard-code one Wave-3 time formula as universally correct. Expose the selected rule set in the debug panel and use relative tolerance consistently.

```

MACD W3 support:
  compare degree MACD behavior at W3 candidate
  record top/highest behavior
  if price advances but MACD structure strongly contradicts W3 source clue, reduce rank / keep
alternate correction candidate
  never let MACD create W3 without valid price structure

```

14. Wave 4 Coding Algorithm

Source basis: Wave 4 may be Zig-Zag, Flat, Triangle, Double or Triple correction. Normal source range is approximately 23.6%-50% using the approved Base-to-W3 convention; terminal cases may allow 61.8%. W4 time is related to W2 and normal impulse overlap rules differ from diagonal/terminal cases.

```

W4_FORMING:
  open correction candidates = {ZigZag, Flat, Double, Triple, Triangle}
  calculate approved Base->W3 retracement
  evaluate W2/W4 alternation and time
  apply overlap rule only after impulse subtype is known

```

The current script uses a 14% normal W4 minimum. Replace that with the approved 23.6% normal minimum; reserve special extension/microscopic handling for the contexts where the notes actually use it.

15. Wave 4 Overlap

Context	Wave-4 / Wave-1 overlap
Normal impulse	Normally invalid according to the normal impulse rule.
Leading Diagonal / terminal context	May be permitted according to diagonal rules.
Ending Diagonal internal structure	Overlap is expected/allowed within the ED rules.
Unknown subtype	Do not hard-invalidate until subtype tests are evaluated.

16. Wave 5 Coding Algorithm

Source basis: Normal W5 uses an impulsive internal structure, 127%-261.8% projection from 3-4 in the source summary, negative MACD divergence vs W3, possible Ending Diagonal, and special truncation/extension cases.

```

W5Projection = abs(P5 - P4) / abs(P3 - P4)

Normal W5 candidate:
  1.27 <= W5Projection <= 2.618
  AND internal impulse valid
  AND W3-not-shortest hard rule passes
  AND W5/W3 divergence context passes configured rule

Also evaluate:
  W5_TRUNCATED
  W5_EXTENDED
  W5_ENDING_DIAGONAL

```

17. Wave 5 Negative Divergence

Compare the oscillator values at the actual degree Wave-3 and Wave-5 pivots. Do not compare random local pivots.

Bullish example:

```
price(P5) > price(P3)
AND MACD(P5) < MACD(P3)
-> negative divergence support
```

Bearish example:

```
price(P5) < price(P3)
AND MACD(P5) > MACD(P3)
-> positive/mirrored divergence support
```

18. Truncated Wave 5

Source basis: When W1 and W3 are both in extension, source notes include a special truncated W5 context with maximum around 81.2% of the 3-4 reference while still containing five sub-moves. Treat as separate subtype, not as a failure of the normal 127% rule.

PART IV - Correction Classifier

19. Why Correction Classification Must Be Candidate-Based

Early A-B price action can legitimately fit more than one correction. Therefore the engine must retain parallel candidates and eliminate them only as rules fail. A manual dropdown that simply changes labels is not a classifier.

After impulse/corrective start:

```
candidates = {ZIGZAG, FLAT, DOUBLE, TRIPLE, TRIANGLE}
```

for each new confirmed subordinate pivot:

```
evaluate each candidate
invalidate impossible candidates
update Fib/time/internal state
if exactly one candidate meets confirmation -> CONFIRMED
if more than one remains -> PRIMARY + ALTERNATE
```

20. Zig-Zag A-B-C

Source basis: Zig-Zag: A-B-C, internal 5-3-5; A fast, B slow, C fast; A minimum around 61.8% of previous-degree W5; B below the Flat-style range; C uses 0-A-B projection and minimum relationship to B; 0-B trendline and correction-end rules support confirmation.

Component	Coding requirement
A	Five-move/impulsive behavior for Zig-Zag context; minimum source retracement.
B	Corrective; generally below 61.8% in the Zig-Zag rule set; detailed developer summary gives roughly 1%-50%.
C	Impulsive five-move candidate; 0-A-B projection; minimum 127% retracement relationship to B in source summary.
Time	B >= A up to selected source limit; C time derives from A/B relationship.
Confirmation	0-B trendline / point-B reversal timing rules where applicable.
End	Whole correction minimum retracement requirement must be evaluated.

21. Flat A-B-C

Source basis: Flat: internal 3-3-5; A minimum around 38.2%; B 61.8%-111%; C behavior determines subtype. Source describes Normal, Elongated, Strong Irregular and Running variants.

```
If B retracement is 61.8%-111% of A:
    Flat candidates become eligible
else:
    do not force Flat just because the label sequence is A-B-C
```

Flat subtype	B context	C context (source summary)
Normal	61.8%-81.2%	C roughly 61.8%-161.8%
Elongated	61.8%-81.2%	C above normal range; source contains max-range conflict
Strong Irregular	61.8%-111%	C roughly 61.8%-161.8%
Running	61.8%-111%	C roughly 161.8%-261.8%

OWNER DECISION / DO NOT GUESS

Wave-C elongated maximum conflicts across the supplied notes (some tables stop at 261.8%, while other pages/graphics extend to 461.8%). Implement the selected conflict mode; do not silently choose.

22. Wave B - Critical Misclassification Rule

Do not use 61.8%-111% as a universal B-wave definition. It belongs to the Flat candidate. Zig-Zag B has a different range. This is the direct coding fix needed to prevent the “B becomes Wave 5 too early” problem from being solved only by a forced label sequence.

```
if B < 0.618 of A:
    keep ZigZag candidate where internal structure supports it

if 0.618 <= B <= 1.11 of A:
    keep Flat candidate

if B > 1.11 of A:
    invalidate/reclassify Flat according to approved rule

Never: “after W3, 2 more pivots occurred, therefore next pivot = W5”
```

23. Wave A

Wave A cannot be defined only as the first pivot after Wave 5. In Zig-Zag context it behaves as a five-move impulsive A; in Flat context it can have a three-move form. It can also form a Leading Diagonal. Use the correction candidate to decide the expected internal structure.

24. Wave C - Completion Must Be Real

Source basis: Wave C: impulsive five moves / 5-3-5-3-5, positive divergence in the source correction rules, 0-A-B extension, minimum 127% retracement of B, normal/truncated/elongated types, A/B-based time rule, and possible Ending Diagonal.

C_FORMING:

- maintain subordinate impulse/ED candidate
- calculate 0-A-B projections
- calculate C-vs-B retracement
- track C time expectation
- track divergence

C_CONFIRMED only when completion conditions pass.

Place C on the actual completion pivot/candle, not simply the third corrective pivot.

PROJECT REQUIREMENT

For this project, the C completion event should be attached to the actual completion candle/pivot when the required Fib objective and completion conditions are achieved. C/Z target events should not be shifted to a later arbitrary swing.

25. Double Correction W-X-Y

Source basis: Double Correction: complex/time-wise, mostly Wave 4; W and X can be Zig-Zag or Flat; Y may be Zig-Zag, Flat or Triangle; small X max 50%, large X 61.8%-111%; Y 61.8%-161.8% of 0-W-X; X time < W; Y time = W, W+X or (W+X)/2; positive MACD divergence at Y is supporting evidence. One source confirmation field is literally unresolved.

DOUBLE_FORMING:

- classify W structure
- classify X as small or large
- validate X time < W
- classify Y
- validate 0-W-X Y projection 61.8%-161.8%
- track MACD divergence near Y completion

If unresolved source confirmation is required:

- status = TBD_BLOCKED or FORMING
- do not invent a rule

26. Triple Correction W-X-Y-XX-Z

Use `XX` internally for the second connecting wave in the Triple Correction. If the chart display must retain the earlier W-X-Y-X-Z wording for familiarity, treat that as a display alias only; the internal data model should distinguish X and XX.

Source basis: Master notes identify Triple W-X-Y-XX-Z and explicitly warn that several source cells are incomplete; later pages supply some XX/Z detail but not every blank field is resolved.

TRIPLE_FORMING:

- W -> X -> Y -> XX -> Z
- store independent durations and Fib relationships
- require X != XX semantic identity even if both are connecting corrections
- evaluate MACD/time toward Y/Z completion
- leave unresolved fields visible as TBD

27. Triangle A-B-C-D-E

Do not label every five-pivot sideways sequence as a triangle. The engine needs geometry, size sequence and boundary checks.

Source basis: Horizontal Contracting Triangle: trendlines travel in opposite price directions; $b < a$, $c < b$, $d < c$, $e < d$; A is widest; thrust after completion is normally 75%-125% of widest wave. Irregular Contracting Triangle has its own B-size condition and c/d/e contraction rules.

```

Contracting triangle candidate:
  legs = {A,B,C,D,E}
  confirm alternating direction
  confirm decreasing leg sizes for subtype
  build upper/lower boundary lines
  validate convergence / source-specific apex rule
  confirm E completion before thrust projection
  thrustTarget = 0.75 to 1.25 * widestLeg (normal source range)

```

Implement expanding triangle separately using its source-specific size sequence. Do not reuse the contracting inequalities in reverse without checking the supplied rule table.

PART V - Diagonals and Extensions

28. Leading Diagonal

Source basis: *Leading Diagonal occurs in Wave 1 and Wave A; source specifies 5-3-5-3-5 framework, special overlap, W3 terminal range, W4 retracement, W5 projection and time behavior.*

```

LD candidate only if parentPosition in {W1, A}
AND internalPattern == 5-3-5-3-5
AND diagonal overlap/range rules pass
AND geometry/trendline behavior passes

```

Do not reject W4/W1 overlap using normal impulse rules until LD is ruled out.

29. Ending Diagonal

Source basis: *Ending Diagonal occurs in Wave 5 and Wave C; source uses 3-3-3-3-3, W4 overlap, W2/W4 restrictions, W5 127%-261.8%, negative divergence from W3 MACD and a post-pattern time expectation.*

```

ED candidate only if parentPosition in {W5, C}
AND internalPattern == 3-3-3-3-3
AND overlap rules pass
AND W2/W4 restrictions pass
AND Fib + divergence support pass

```

30. Extension Engine

Source basis: *Extension move counts use 5, 9, 13, 17, 21; maximum 21. Every extension subtype changes follow-on retracement/structure expectations. W1, W3 and W5 may extend, with W3 being the common case.*

```

extensionMoveCount in {5,9,13,17,21}

if internalCount > 5:
  classify parent as extension candidate
  apply subtype-specific rules
  do not let every internal pivot advance parent wave number

```

This module is one of the main reasons raw pivots and main-degree pivots must be separate.

PART VI - Fibonacci, Time, MACD, Trendlines and Project Context

31. Wave-Specific Fibonacci Anchors

Candidate	Required anchor idea
Wave 2	Point 0 -> Wave 1 retracement

Candidate	Required anchor idea
Wave 3	0-1-2 extension
Wave 4	Approved Base -> W3 source convention
Wave 5	3-4 projection
Wave C	0-A-B extension / B retracement relationship
Wave Y	0-W-X extension
Wave Z	Triple-correction source anchors; expose selected formula in debug
Degree W1	Important High/Low range

Draw only levels relevant to the active candidate. Label the anchor pair. A generic Important H/L ladder is useful as a guide but is not a substitute for wave-specific Fibonacci logic.

32. Source Fibonacci Universe

The supplied notes use levels including 14/14.2, 23.2/23.6, 38.2, 50, 61.8, 81.2, 100, 111, 127/127.2, 161.8, 200, 261.8, 361.8 and 461.8, with larger W3 extension values in special cases. Do not display all levels at once; activate by candidate.

33. Time Engine

Time rules must be part of candidate evaluation, not only a dashboard note. Use duration from actual confirmed pivot times/bars. Preserve source conflicts.

Structure	Time relationship to implement
Wave 2	Source variants include 1/4, 1/2, 1X, 2X of W1; exact allowed set is conflict-sensitive.
Wave 3	Source variants derive from W1/W2; implement selected rule and show it.
Wave 4	Related to W2; source versions differ (e.g., 1/2, 2X, 3X in one summary; other notes show 5X). W4 should not simply equal W2 in the relevant rule set.
Wave B	$B > A$; subtype-dependent upper timing context.
Wave C	If $A=B$, $C=A+B$; if $A<B$ /different, source uses an average formula in the developer summary.
Wave X	$X < W$.
Wave Y	$Y = W$, $W+X$ or $(W+X)/2$.
Wave XX/Z	Use source-specific Triple Correction relationships; do not invent missing rows.

```
relativeTimeError = abs(actual - expected) / expected
timePass = relativeTimeError <= configuredTolerance
```

Use relative tolerance consistently. The current script mixes relative and absolute ratio tolerance in different places.

34. Wave-Specific MACD / RSI

Stage	MACD / oscillator role
Base / W1	Lowest low or positive divergence support.
W2	Falls slowly; should respect approved base-MACD behavior.
W3	Top/highest momentum clue.
W4	Can fall sharply / form low reading in price-wise correction.
W5	Negative divergence vs W3.
Simple C	Positive divergence can support completion.
B / X	MACD can move with speed; advanced classifier uses high MACD without matching price extension as clue.
Y / Z	Positive divergence may support complex-correction completion.
RSI	Project supporting divergence; use corresponding Elliott pivots, not random local swings.

Keep MACD fast/slow/signal and divergence pivot sensitivity configurable because the source does not specify exact numerical parameters. Also expose whether divergence is computed from MACD line, histogram or both.

35. Trendline / Channel Engine

Support the Elliott-specific lines requested by the source: 0-2, 1-3, 2-4, 0-B and triangle boundaries. Store line anchors and break time/state. These lines are part of confirmation and should not be decorative only.

36. FBD / FBO Controlled Recount

PROJECT REQUIREMENT

FBD/FBO is a structural context trigger for this project. It should nominate a possible new Elliott origin, not directly assign a new wave count.

```
onConfirmedFBDorFBO(event):
    candidateOrigin = event.structuralPivot
    if candidateOrigin passes important/degree tests:
        compare with current locked base
        if current count hard-invalid OR candidate clearly superior by approved rule:
            start controlled alternate/recount
        else:
            keep current count; store event as supporting context
```

37. Degree / Multi-Timeframe Handling

Every candidate must store its degree and source timeframe. The source documents contain a default Daily / 75m / 15m degree framework, while the broader project may use a richer timeframe ladder. Implement degree mapping as a configurable routing layer; do not let a 5-minute internal pivot and a Daily major pivot live in the same wave array without degree metadata.

PROJECT REQUIREMENT

The project requires multi-timeframe Elliott degrees to remain distinct. A lower-timeframe sub-wave can explain the internal structure of a higher-degree wave, but must not automatically overwrite the higher-degree count.

PART VII - Live Behavior, Recount and Non-Repaint

38. Developing vs Confirmed Pivots

The most recent raw pivot may continue to move to a more extreme price before its reversal is confirmed. That is acceptable for a FORMING/DEV wave. It is not acceptable to visually present that pivot as permanently confirmed and later move it without explanation.

FORMING label -> may update
 CONFIRMED label -> locked

A confirmed label changes only after an approved hard structural invalidation / controlled recount event.

39. Controlled Recount Rules

Trigger	Allowed behavior
Minor new raw swing	Do not restart main count.
Hard wave invalidation	Reject candidate and promote valid alternate / recount.
New valid Important H/L base	Open alternate origin; do not instantly replace locked base.
Confirmed FBD/FBO important reversal	Open structural recount candidate.
Correction resolves differently	Promote surviving correction candidate.
Higher-degree completion / degree fall	Start new degree candidate according to routing rules.

40. Alternate Count

Primary: W4 FORMING - Flat candidate
 Alternate: Wave B FORMING - larger correction candidate

Keep both until:
 one violates a hard rule, OR
 one reaches its confirmation conditions.

Do not force Wave 5 simply because it is the next expected label after a fixed number of pivots.

PART VIII - Exact Updates Required in the Current Pine Script

Current element	Keep / replace	Required update
ta.pivohigh / ta.pivotlow infrastructure	KEEP	Use as raw confirmed pivot provider; preserve confirmation delay semantics.
ATR / minimum swing filters	KEEP	Use for raw swing quality only, not Elliott semantics.
swing arrays storing MACD/RSI	KEEP	Extend with degree, status, parent/child and candidate IDs.
f_phase / cycleLength / modulo logic	REPLACE	Remove as primary wave-number engine.
f_valid_anchor_index rescanning	REPLACE	Use persistent base candidate / locked base lifecycle.

Current element	Keep / replace	Required update
f_rule_state warnings	REFACTOR	Convert hard rules into candidate rejection; soft rules into ranking/debug evidence.
Wave 2 min 14%	FIX	Normal W2 min 23.6%; 14% separate microscopic subtype.
Wave 2 no explicit 81.2 normal max	FIX	Implement normal max/context and owner-selected invalidation behavior.
Flat B 61.8-111 applied generically	FIX	Apply only inside Flat candidate.
showWave4Internal forced A/B/C sequence	REPLACE	Use actual correction candidates; do not force positional A/B/C before W5.
Manual correctionPattern dropdown	KEEP AS DISPLAY FILTER ONLY	Detection must be automatic.
Wave 4 min 14%	FIX	Use approved normal 23.6% source range.
W4 overlap universal	FIX	Make subtype-aware; diagonals/terminal exceptions.
W5 127-261.8 target infrastructure	KEEP	Integrate into proper W5 classifier plus truncation/extension/ED.
Generic Important H/L fib ladder	KEEP AS GUIDE	Add wave-specific Fib engines.
Time table	KEEP	Refactor into actual candidate constraints; show selected conflict mode.
Drawing cleanup arrays	KEEP	Reuse for labels/lines, but attach objects to candidate IDs/status.
No Leading/Ending Diagonal engine	ADD	Implement dedicated subtype detectors.
No extension move-count engine	ADD	5/9/13/17/21 internal count classifier.
No WXY/Triple structural engine	ADD	Implement pattern rules instead of text labels.
No controlled FBD/FBO recount	ADD	Project requirement.
No HP BUY/SELL	ADD	Project Wave-2 signal with confirmation.
No Pine/Python parity output	ADD	Same OHLC must produce materially same confirmed count.

PART IX - Suggested Engine Pseudocode

41. Main Event Loop

```
onBarClose():
    updateIndicators()
    newPivot = detectConfirmedRawPivot()

    if newPivot:
        rawSwings.upsert(newPivot)
        updateImportantHLAndDegreeContext()
        updateBaseCandidates(newPivot)

        if lockedBase exists:
            updateImpulseCandidates(newPivot)
            updateCorrectionCandidates(newPivot)
            updateAlternateCounts(newPivot)

        applyHardInvalidations()
        promoteConfirmedCandidates()
        performControlledRecountIfRequired()
        updateDrawingsFromCandidateState()
        updateDebugPanel()
```

42. Candidate Validation Order

```
validate(candidate):
    if not parentContextAllows(candidate): INVALID
    if hardPriceRuleFails(candidate): INVALID
    if internalStructureImpossible(candidate): INVALID

    calculateFib(candidate)
    calculateTime(candidate)
    calculateMomentumSupport(candidate)
    calculateTrendlineSupport(candidate)

    if mandatorySourceConditionsComplete:
        CONFIRMED
    else:
        FORMING

    return status + exact reason codes
```

43. Example Reason Codes

Code	Meaning
W2_ORIGIN_BREAK	Wave 2 crossed Point 0.
W2_RETRACE_OUTSIDE_NORMAL	W2 outside normal source retracement range.
W3_BELOW_TRENDING_MIN	W3 below 161.8%; terminal classifier must decide.
W3_INTERNAL_FAIL	Expected impulsive internal structure failed.
W4_NORMAL_OVERLAP	Normal impulse W4 entered W1 territory.
W5_W3_SHORTEST	Proposed W5 makes W3 the shortest impulse wave.

Code	Meaning
FLAT_B_GT_111	Flat B exceeded approved maximum.
C_INTERNAL_INCOMPLETE	C Fib reached but internal completion missing.
DOUBLE_CONFIRM_TBD	Source confirmation field unresolved.
BASE_REPLACEMENT_NOT_PROVEN	New origin candidate is not strong enough to replace locked base.

PART X - Acceptance Test Matrix

ID	Module	Test condition	Expected result
T01	Base/W1	Important H/L + base oscillator evidence + 61.8% degree progress	W1 FORMING -> CONFIRMED; base locked.
T02	Normal W2	23.6%-81.2% retracement with valid correction	W2 remains valid; Point 0 preserved.
T03	Microscopic W2	Special ~14% case	W2_MICROSCOPIC; W3 extension candidate.
T04	Invalid W2	W2 crosses Point 0	Reject W2/impulse; controlled recount.
T05	HP BUY	Bull W1 + W2 61.8%-81.2% + correction completion + confirmation	HP BUY only after confirmation.
T06	HP SELL	Bearish mirror	HP SELL only after confirmation.
T07	Trending W3	Internal impulse + >=161.8% + MACD support	W3_TRENDING CONFIRMED.
T08	Terminal W3	Valid terminal structure below 161.8%	W3_TERMINAL; not generic warning.
T09	Extended W3	9/13/17/21 internal moves	Extension subtype; main count remains W3.
T10	Normal W4	Valid 23.6%-50% correction	W4 classification and time/alternation visible.
T11	Complex W4	Double/Triple/Triangle example	No premature W5.
T12	Normal W5	127%-261.8% 3-4 + internal + divergence	W5 CONFIRMED.
T13	Truncated W5	Valid double-extension context	W5_TRUNCATED, not rejected by normal min.
T14	Leading Diagonal	Valid W1/A LD	LD subtype; overlap allowed by LD rules.
T15	Ending Diagonal	Valid W5/C ED	ED subtype + divergence/post-pattern logic.
T16	Zig-Zag	5-3-5 + correct A/B/C	Automatic Zig-Zag classification.

ID	Module	Test condition	Expected result
T17	Flat	3-3-5 + B 61.8%-111%	Correct Flat subtype.
T18	B below Flat range	B <61.8 with valid Zig-Zag context	Do not force Flat.
T19	B >111	Flat invalidation case	Flat candidate rejected/reclassified.
T20	Wave C	C internal impulse + Fib/time/divergence	C on actual completion pivot/candle.
T21	Double WXY	Small/large X and valid Y projection/time	Automatic Double candidate; unresolved confirmation marked TBD.
T22	Triple	W-X-Y-XX-Z	All components and time/Fib relationships tracked.
T23	Contracting Triangle	ABCDE with correct geometry/size sequence	Triangle subtype + thrust projection.
T24	Expanding Triangle	Source-valid expanding case	Separate expanding subtype.
T25	False small swings	Many internal pivots inside W3	Main count remains W3.
T26	FBD/FBO recount	Important confirmed reversal event	Open recount candidate; no blind reset.
T27	Alternate count	Ambiguous W4 vs B / Flat vs Zig-Zag	Primary + alternate until resolution.
T28	Non-repaint	Live sequence with developing last leg	Only FORMING pivot moves; confirmed labels stay locked.
T29	144-day test	Daily, 1H, 15m	Same intended Important H/L source context, not 144 local bars.
T30	Pine/Python parity	Same OHLC dataset	Materially same confirmed count; differences documented.

44. Mandatory Developer Evidence for Each Test

- Chart screenshot with Elliott labels and candidate status.
- Exact raw pivots used for the main-degree count.
- Numeric Fib ratios and anchors.
- Wave durations and selected time rule.
- MACD/RSI values at compared pivots where divergence is used.
- First hard invalidation reason if a candidate is rejected.
- Primary vs alternate candidate where applicable.
- Pine output and Python output for parity cases.

PART XI - Source Conflict Register / Owner Decisions

ID	Topic	Conflict / uncertainty	Developer action
C01	Wave-2 time	Source pages give different allowed time relationships.	Configurable conflict mode; do not silently choose.

ID	Topic	Conflict / uncertainty	Developer action
C02	W4 max during W1 extension	38.2% vs 50% appears in different sections/examples.	Keep context-specific setting until owner locks.
C03	W3 terminal/trending boundary	Main degree min 161.8%; terminal subordinate may begin near 61.8% and stay below 161.8%.	Explicit classifier; define equality/tolerance.
C04	LD/ED coexistence wording	Source wording is ambiguous about coexistence in same impulse.	Owner decision / visible mode.
C05	Zig-Zag / Wave-C max	261.8% vs 461.8% appears across tables/graphics.	Do not auto-reconcile.
C06	Double correction confirmation	Source cell is "???".	TBD-BLOCKED / FORMING; do not invent.
C07	Double/Triple move totals	Summary totals differ by pattern combination.	Implement per-pattern tables and log combination.
C08	Small vs large X	Different general/specific ranges appear.	Classify small vs large X explicitly.
C09	Triple X/XX time	Different X-time constraints appear.	Context-specific rule + owner choice.
C10	Z failure / failure percentage	Different nearby notes reference 31.8 vs 38.2 and different wave labels.	Do not hard-code until confirmed.
C11	MACD parameters	No exact fast/slow/signal or divergence algorithm in source.	Inputs + visible settings.
C12	Pivot definition	No single exact wave endpoint algorithm in source.	Owner-approved pivot engine; do not let implementation silently redefine theory.
C13	Qualitative terms	Fast/slow/steep/microscopic/not long are not fully quantified.	Keep non-blocking or expose thresholds until approved.

PART XII - Required Implementation Sequence

Phase	Deliverable	Acceptance gate
Phase 1 - P0	State engine, raw/main pivot separation, locked base, hard invalidation, 144-day fix, controlled recount skeleton	T01, T04, T25, T28, T29 must pass before proceeding.
Phase 2 - Core impulse	W1/W2/W3/W4/W5 + HP + normal Fib/time	T02-T13.
Phase 3 - Corrections	Zig-Zag, Flat, A/B/C, WXY, Triple, Triangle	T16-T24.
Phase 4 - Special impulse	Leading/Ending Diagonal + extensions/truncation	T09, T13-T15.
Phase 5 - Supporting engines	MACD/RSI, trendlines/channels, FBD/FBO recount, degree routing	T26-T27 plus debug evidence.

Phase	Deliverable	Acceptance gate
Phase 6 - Parity/production	Python parity, dashboard/debug, performance optimization	T30 plus full regression suite.

45. What the Developer Must Return

1. Updated Pine Script source.
2. Updated Python implementation using the same rule/state model.
3. Implementation checklist: DONE / PARTIAL / TBD for every numbered requirement in this manual.
4. Visible list of all owner decisions / source conflicts still unresolved.
5. Screenshots and debug records for the acceptance tests.
6. Pine-vs-Python comparison on identical OHLC data.
7. Non-repaint statement explaining what is mutable (FORMING) and what is locked (CONFIRMED).
8. Performance note: maximum raw pivots, candidate limits, request.security usage and drawing-object limits.

46. Definition of Done

The Elliott module is complete only when...

The engine can explain why Point 0 was selected; why W1/W2/W3/W4/W5 are structurally valid; which correction type is active; why a B move is not prematurely W5; where C actually completed; which Fib anchors and time rule were used; what invalidates the count; why a recount occurred; which labels are locked; and whether Pine and Python agree.

Do not accept this shortcut

"The chart displays 1-2-3-4-5-A-B-C" is NOT proof that Elliott Wave logic is implemented. The labels must be the output of the rules, not the input assumption.

Appendix A - Quick Formula Reference

Formula / concept	Implementation
Bull W2 retracement	$(P1 - P2) / (P1 - P0)$
Bear W2 retracement	$(P2 - P1) / (P0 - P1)$
W5 projection base	$\text{abs}(P3-P4)$; W5 ratio = $\text{abs}(P5-P4)/\text{abs}(P3-P4)$
Flat B retracement	$\text{abs}(B-A) / \text{abs}(A-\text{startOfA})$, using exact candidate anchors
Relative time tolerance	$\text{abs}(\text{actual-expected})/\text{expected} \leq \text{tolerance}$
Bull W5 negative divergence	$\text{price5} > \text{price3}$ AND $\text{oscillator5} < \text{oscillator3}$
Bear W5 mirrored divergence	$\text{price5} < \text{price3}$ AND $\text{oscillator5} > \text{oscillator3}$
Contracting triangle basic size order	$A > B > C > D > E$ for horizontal contracting subtype
Normal HP zone	W2 retracement 61.8%-81.2% + correction completion + project confirmation

Appendix B - Developer Checklist Before Coding

- ☐ Have you removed the modulo/fixed-cycle count as the primary wave engine?
- ☐ Are raw swings and Elliott pivots stored separately?
- ☐ Can a hard-invalid candidate be rejected instead of merely colored red?
- ☐ Is the confirmed base persistent/locked?

- ☐ Is Important H/L truly degree/timeframe-aware rather than 144 local bars?
- ☐ Does W2 use 23.6%-81.2% as normal range and keep 14% as a special microscopic case?
- ☐ Does the Flat B 61.8%-111% rule apply only to Flat candidates?
- ☐ Can the engine keep Zig-Zag and Flat as simultaneous early candidates?
- ☐ Does C wait for actual completion instead of fixed pivot position?
- ☐ Can W3 be Trending, Terminal or Extended?
- ☐ Can W5 be Normal, Truncated, Extended or Ending Diagonal?
- ☐ Are Leading/Ending Diagonal overlap rules separated from normal impulse rules?
- ☐ Does extension counting use internal 5/9/13/17/21 logic?
- ☐ Can W4 automatically become Zig-Zag/Flat/Double/Triple/Triangle?
- ☐ Is Triple Correction stored as W-X-Y-XX-Z internally?
- ☐ Are Fib anchors wave-specific and shown in debug?
- ☐ Are time-rule source conflicts visible/configurable?
- ☐ Are MACD parameters/divergence settings configurable?
- ☐ Can FBD/FBO nominate a recount without blindly resetting?
- ☐ Are FORMING and CONFIRMED labels visually/state-wise different?
- ☐ Can the engine maintain an ALTERNATE count?
- ☐ Do Pine and Python return materially the same confirmed count?

Appendix C - Source Basis Used for This Manual

- Current Pine Script supplied for audit: "Elliott Wave Notes Overlay" (1,007-line pasted Pine Script).
- EW Notes / Chartking v2 reference material: impulse Waves 1-5, diagonals, extensions, Zig-Zag, Flat, Double, Triple, Triangle, MACD, trendlines/channels and time rules.
- Elliott Wave Indicator - Developer Requirement Specification: normalized rule tables, candidate-state architecture, conflict register and acceptance matrix.
- Elliott Wave Master Notes / Pivot Table: consolidated pivot-by-pivot quick reference and unresolved-source warnings.
- Project-specific requirements carried into this handoff: HP BUY/SELL at deep Wave-2 completion with confirmation; FBD/FBO controlled recount; actual C/Z completion-event marking; degree-aware multi-timeframe separation; Pine/Python parity.

END OF DEVELOPER CODING MANUAL

DETAILED IMPLEMENTATION ADDENDUM - FOR A PINE SCRIPT EXPERT WITH ZERO ELLIOTT WAVE BACKGROUND

Why this addendum exists

The earlier manual gives the architecture and missing modules. This addendum goes one level deeper: it explains how a programmer should think about each Elliott object, which prices become anchors, how to calculate ratios, when a candidate is only FORMING, what confirms it, what invalidates it, how to preserve source conflicts, and exactly how the current Pine overlay should be refactored.

Developer mindset

Treat Elliott Wave as a constrained parser over confirmed price pivots, not as a label sequence. The algorithm should parse raw pivots into several possible structures, reject impossible structures, rank the remaining structures using Fib/time/MACD/trendline evidence, and only then print confirmed labels.

A. What You Must Understand Before Writing Any More Pine Code

A1. There Are Three Different Things Called a “Swing”

Object	What it is	Can it advance main Elliott count?	Can it move later?
Raw pivot	A confirmed local high/low from pivot/ATR logic.	No. It is only input data.	The newest same-type raw pivot may be replaced until its reversal is confirmed.
Internal/subordinate pivot	A raw pivot selected as part of a larger wave internal structure.	No, unless it also qualifies at the parent degree.	Only while parent/internal leg is FORMING.
Main-degree Elliott pivot	A pivot that completed the required structure/Fib/time conditions for the selected degree.	Yes.	Normally locked once CONFIRMED; only controlled recount can change it.

IMPLEMENTATION WARNING

The current script does not make this separation strongly enough. Its accepted swing sequence directly drives the wave phase. That is the root cause of small swings becoming Elliott labels too early.

A2. The Correct Question at Every New Pivot

WRONG QUESTION:

“What label comes after the last label?”

CORRECT QUESTIONS:

1. Which currently FORMING candidates can this pivot legally complete?
2. Which candidates become impossible because of this pivot?
3. Does the pivot belong to the current degree or only to an internal degree?
4. Do Fib/time/internal rules confirm anything now?
5. If multiple candidates remain, which is PRIMARY and which is ALTERNATE?

A3. Impulse Rules vs Correction Guidelines

The source explicitly distinguishes impulse analysis as rule-based and correction analysis as guideline-based. In software terms, this means impulse has stronger hard-invalid conditions, while corrections need parallel candidates because the same early A-B path can later become Zig-Zag, Flat, Double, Triple or Triangle.

Source basis: *Developer Requirement Specification p.2; Charting EW Notes p.3.*

B. Exact Audit of the Current 1,007-Line Pine Script

The following line ranges refer to the supplied file "Elliott Wave Notes Overlay". The aim is to tell the developer what can be reused and what must be replaced.

Current line range / function	What it currently does	Decision	Exact update
147-197: f_correction_count / f_cycle_length / f_phase / f_phase_text	Builds a fixed repeating sequence of 0,1,2,3,A,B,4/C,5 and correction labels.	REPLACE AS COUNT ENGINE	Keep only optional text-format helpers. Wave identity must come from candidate validation, never modulo phase.
257-321: f_start_confirmation_note	Important H/L, ATR, degree, MACD extreme and RSI/MACD divergence checks.	KEEP + REFACTOR	Split into BaseCandidate scoring and BaseConfirmation. Make oscillator mode explicit: extreme OR divergence is not the same as mandatory divergence.
323-353: f_valid_anchor_index / f_wave_phase	Rescans and lets the latest passing anchor determine cycle phase.	REPLACE	Persist BASE_CANDIDATE / BASE_LOCKED. A new passing base opens an alternate origin; it does not automatically replace the locked base.
355-483: swing filters / f_upsert_swing	Maintains filtered alternating pivots and replaces same-type pivots with more extreme ones.	KEEP AS RAW PIVOT ENGINE	Rename conceptually to rawSwings. Add pivot confirmation state and do not attach Elliott labels directly.
485-689: f_rule_state	Calculates warning/invalidation booleans after phase label is already known.	REFACTOR	Each wave validator should return CandidateStatus. Hard invalid must reject the candidate; soft warning should only reduce rank/show debug.
509-537: W2	Origin invalidation + minimum retrace/time warning.	FIX	Normal W2 range depends on selected source mode; implement explicit microscopic subtype and max-range handling.
539-579: W3	161.8% ratio + one time formula.	EXPAND	Add internal 5-move/extension count, Trending vs Terminal subtype, MACD top support, source-selected time rule.
581-657: W4 A/B/C and W4	Forces internal A/B/C by pivot position and uses Flat B range.	REPLACE	Create real correction candidates. Do not force every W4 to be A-B-C or every B to use 61.8%-111%.
659-687: W5	Normal W5 127%-261.8 + W3-shortest invalidation.	KEEP + EXPAND	Add W5 divergence, truncation, W5 extension, Ending Diagonal, internal move count.
691-797: time table	Displays selected time relationships.	KEEP UI + REWRITE LOGIC	Make time rules part of candidate evaluation and show which source mode is active.
819-887: Fib guide + W5 targets	Important H/L Fib ladder and W5 projection.	KEEP DRAWING INFRA	Add wave-specific anchor sets: 0-1,0-1-2,0-3,3-4,0-A-B,0-W-X,0-Y-XX.
891-947: redraw	Draws every phase label even if hardInvalid.	REPLACE DRAW DECISION	FORMING candidates may be optional/dotted; CONFIRMED labels locked; INVALID candidates should not look like accepted waves.

Current line range / function	What it currently does	Decision	Exact update
949-1007: pivot calculation	Confirmed pivots with right-bar delay.	KEEP	Document delay as confirmation latency, not repainting; attach pivot to raw degree buffer only.

C. Foundation Math Helpers the Developer Should Build First

Before implementing any pattern, create a small library of direction-safe math functions. This reduces long/short asymmetry bugs.

C1. Direction Convention

```
direction = +1 for bullish candidate
direction = -1 for bearish candidate

moveSigned(a, b, direction) = direction * (b - a)
moveAbs(a, b) = abs(b - a)
```

C2. Retracement and Extension Functions

Generic retracement	$\text{retr} = \text{abs}(\text{retraceEnd} - \text{retraceStart}) / \text{abs}(\text{referenceEnd} - \text{referenceStart})$
Bull W2	$(P1 - P2) / (P1 - P0)$
Bear W2	$(P2 - P1) / (P0 - P1)$
W3 0-1-2 extension ratio	$\text{direction} * (P3 - P2) / \text{abs}(P1 - P0)$
W4 Base-to-W3 retracement	$\text{abs}(P3 - P4) / \text{abs}(P3 - P0)$ [use only if this is the owner-approved anchor convention]
W5 3-4 projection ratio	$\text{direction} * (P5 - P4) / \text{abs}(P3 - P4)$
ABC 0-A-B C extension	$\text{directionOfA} * (C - B) / \text{abs}(A - \text{originA})$
C retracement of B	$\text{abs}(C - B) / \text{abs}(B - A)$
WXY Y extension	$\text{directionOfW} * (Y - X) / \text{abs}(W - \text{originW})$
Triple Z extension	$\text{directionOfY} * (Z - XX) / \text{abs}(Y - \text{originTriple})$

Important

The formulas above are coding translations of the source Fib anchor descriptions. The debug panel must always print the anchor prices used. If the owner chooses a different source convention for a conflicted anchor, change the helper input - not scattered pattern code.

C3. Time Helpers

Duration	<code>endBar - startBar</code> (or exact timestamp difference when required)
Relative tolerance	<code>abs(actual - expected) / max(expected, eps) <= tolerancePct</code>
Absolute bar tolerance	<code>abs(actualBars - expectedBars) <= toleranceBars</code>

IMPLEMENTATION WARNING

Do not compare 0.5x time using a raw absolute tolerance while comparing another rule using percentage tolerance. The current script mixes these styles. Standardize the calculation and expose the tolerance source.

C4. Divergence Helper

Bullish positive divergence:
`price2 < price1 AND oscillator2 > oscillator1`

Bearish negative divergence:
`price2 > price1 AND oscillator2 < oscillator1`

For W5 vs W3, compare only the actual degree W3 and W5 pivots.
 For base divergence, compare same-type important lows/highs at the selected degree.

D. Base Point and Degree Formation - Step-by-Step

D1. Timeframe and Degree Are Configuration, Not One Universal Mapping

The supplied notes contain more than one timeframe mapping. Chartking uses a Daily/75m/15m framework for Major/Intermediate/Minor in its main notes, while the Hardik material changes degree timeframes depending on investment, swing or day-trading horizon. Do not silently combine them.

Source	Major	Intermediate	Minor / lower	Implementation
Chartking main framework	Daily	75-minute	15-minute	Use as one selectable preset.
Hardik - 1 year holding	Monthly	Weekly	Daily	Selectable long-horizon preset.
Hardik - swing 1-3 months	Weekly	Daily	75m / 288m / 1h commodity	Selectable swing preset.
Hardik - day trading	Daily	75m	15m	Essentially matches Chartking day-trading mapping.

SOURCE CONFLICT - DO NOT GUESS

Timeframe/degree mapping differs across the supplied notes. Add a "Degree Mapping Preset" input and show the active mapping in the dashboard. Never let timeframe routing change silently because current price crosses an arbitrary threshold unless that source mode is explicitly enabled.

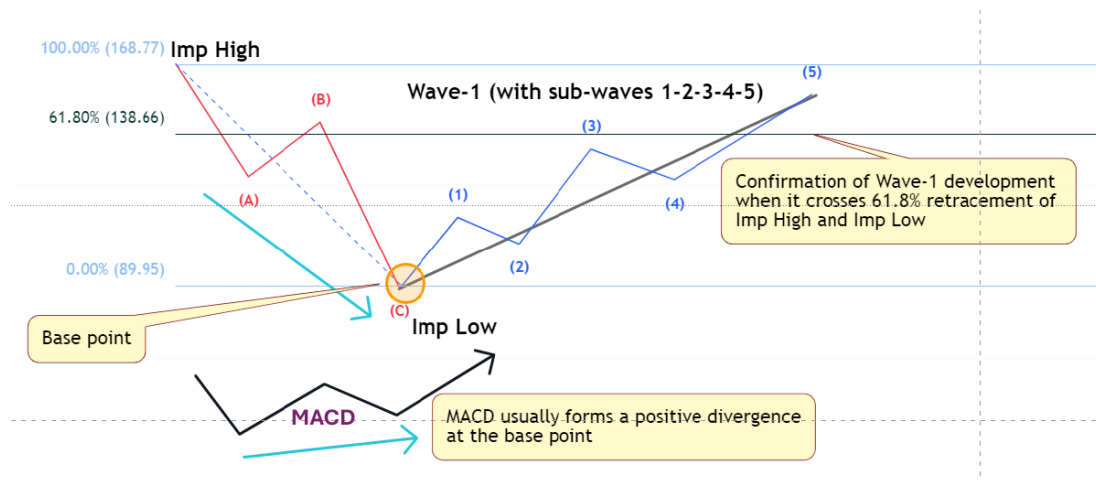
D2. Base Search Workflow

- Choose the analysis/degree timeframe preset.
- Build Important High and Important Low for the approved lookback. Chartking says max 144 days for the Daily start search; Hardik also requires significant high/low context.
- Find candidate Point 0 near a MACD lowest point and/or positive divergence according to the selected oscillator mode.
- Store both bullish and bearish base candidates until structure chooses one; do not assume direction too early.
- Do not CONFIRM Wave 1 merely because Point 0 exists. Point 0 is only the origin candidate.

Source basis: Chartking EW Notes pp.3-6; Hardik pp.8-10; Developer Specification pp.3-5.

[Index](#)

Development of Wave-1



D3. 61.8% Degree Confirmation

The source uses the move from Important Low toward Important High (or the bearish mirror) to determine whether the developing move has reached degree scale. The developer must separate “five internal moves completed” from “degree Wave 1 confirmed”. If five moves finish before the degree threshold, the notes may treat them as subordinate rather than the full degree wave.

```
if fiveInternalMovesCompleted AND degreeProgress < selectedDegreeThreshold:
    classify structure as SUBORDINATE candidate
    do not advance parent degree to W1 confirmed

if degreeProgress >= selectedDegreeThreshold AND internalStructureValid:
    W1 can move from FORMING -> CONFIRMED
```

D4. 144 Days Is Not 144 Bars

```
// WRONG on an intraday chart
importantHigh = ta.highest(high, 144)

// CORRECT concept
importantHigh = request.security(symbol, approvedDegreeTf, highest(high, barsEquivalentToSourceWindow))
// or use a timestamp-based 144-day window if the source says calendar days.
```

IMPLEMENTATION WARNING

The current script input “Important H/L lookback bars = 144” changes meaning with chart timeframe. This must be fixed before the wave-count logic can be trusted across timeframes.

E. Wave 1 - Detailed Theory-to-Code Translation

E1. What Wave 1 Means

Wave 1 is not simply "the first higher high after a low." It is the first directional degree candidate after the validated base. The source describes it as impulsive with five subordinate movements, with MACD base evidence and degree confirmation from the Important H/L range.

Requirement	What the developer must store/check
Origin	Locked base Point 0, price/time/degree.
Internal movement	At least a valid five-move impulse candidate, unless a specifically allowed diagonal subtype is being parsed.
Degree progress	Progress from Important Low/High to the source 61.8% degree threshold.
MACD support	Lowest low or positive divergence at the base according to selected oscillator configuration.
Channel	After the source threshold, channel/trendline module can activate.
Status	FORMING before all requirements; CONFIRMED only after source conditions.

E2. Bullish Wave 1 Pseudocode

```

evaluateBullW1(base, rawPivots):
    candidate = buildFiveMoveImpulseFrom(base, rawPivots)
    candidate.degreeProgress = (candidate.endPrice - importantLow) / (importantHigh - importantLow)

    if candidate.breaksBase:
        return INVALID("W1_BASE_BREAK")

    if not candidate.internalFiveMovePossible:
        return FORMING or INVALID depending on structure

    if candidate.degreeProgress >= selectedW1DegreeThreshold:
        return CONFIRMED

    return FORMING("SUBORDINATE_OR_DEGREE_NOT_REACHED")

```

E3. Wave 1 Time - Source Variants

The Hardik document gives an additional time method: measure bars from significant high to significant low, divide by 50 for a maximum Wave-1 time reference, and also describes a 50% price recovery within that time as sufficient when five moves exist. The separate Time Analysis document instead describes 61.8% recovery in about half the previous decline time or 100% recovery in about the same time. These are not identical rules.

Oct 02, 2025 15:36 UTC+5:30

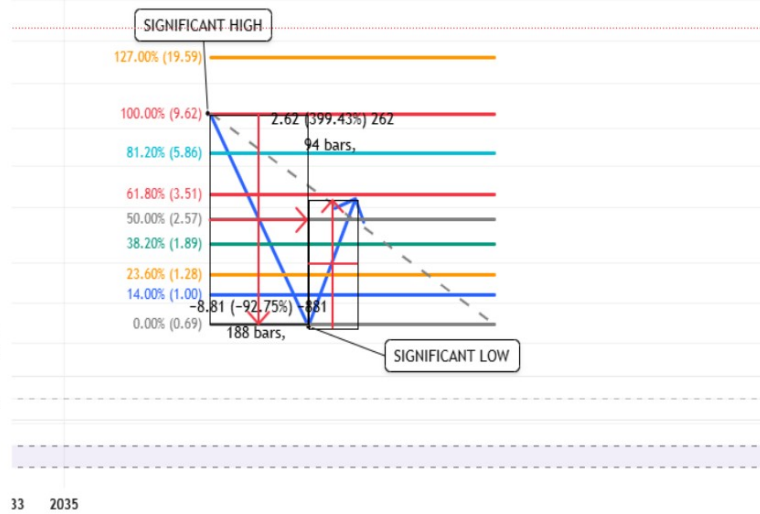
3.31 -0.38 (-1.32%)

10.50 0

Time Analysis

➤ Wave 1

- Time of wave 1 can be measured by putting date and time measurement tool from significant high to significant low.
- Now divide no. of bars by 50.
- And then again put tool from base point to the no. of bars which we have derived from dividing with 50.
- This is maximum time taken by wave 1 to complete. Remember it can also take less time than max time.
- If price reaches fib retracement till 50% in a given time period then also we can consider as wave 1 (only if we get 5 moves)

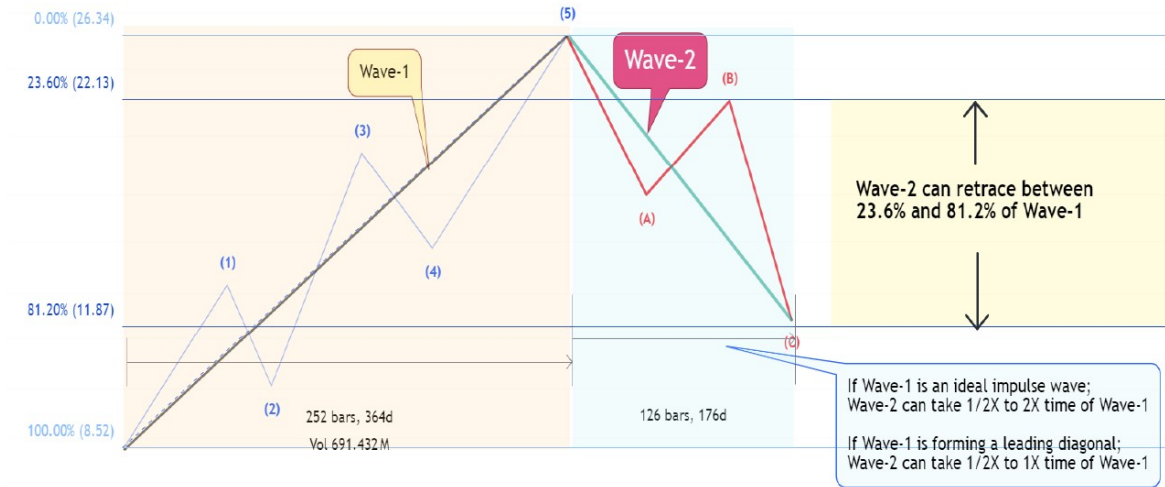


Source visual: Hardik Wave-1 time method using significant High-Low measurement (Hardik p.14).

SOURCE CONFLICT - DO NOT GUESS

Wave-1 time has multiple source methodologies. Implement them as separate Time Rule presets (e.g., CHARTKING/HARDIK/TIME_ANALYSIS) rather than averaging them into a new rule.

F. Wave 2 - Detailed Correction and Trade Logic

[Index](#)


Source visual: Wave-2 retracement zone and time context (Charting EW Notes p.8).

F1. Why the Current 14% Rule Is Dangerous

One source table says normal W2 minimum 23.6% with a separate microscopic 14% clue. Other supplied impulse/Hardik notes describe the minimum as 14% or 14%-23.6%. Therefore 14% cannot be silently treated as a single universal meaning.

Source reading	14% meaning	23.6% meaning	Implementation
Chartking primary table	Microscopic special case that often implies W3 extension	Normal W2 minimum	Recommended "Chartking" mode.
Hardik degree notes	Minimum can begin 14%-23.6%	Also an important minimum zone	Alternative source preset.
Impulse Notes	W2 minimum stated as 14%	Included as part of Fib ladder	Alternative legacy preset.

SOURCE CONFLICT - DO NOT GUESS

Add W2MinimumMode. Do not "fix" the source by choosing one rule invisibly. The dashboard/debug log must show whether the active W2 candidate is NORMAL, MICROSCOPIC, or LEGACY_14_MIN.

F2. W2 Structural Sequence

```

After W1 confirmation:
  open W2 correction container
  allowed child candidates = ZigZag, Flat, WXY, approved single-leg case
  
```

A raw low is NOT W2 simply because price retraced.
W2 becomes CONFIRMED only after one child correction candidate completes
and Point 0/base is still structurally valid.

F3. W2 Range and Invalidation

Event	State effect
Retracement below first source threshold	W2 FORMING / microscopic possibility, depending mode.
Retracement inside selected normal range	Valid W2 price context, but not yet confirmed without internal completion.
Retracement 61.8%-81.2%	Deep W2 context; project HP zone becomes eligible.
Retracement beyond 81.2%	Flag according to selected source mode; keep base-origin hard rule separate.
Price crosses Point 0	Hard invalidation of the proposed impulse; reject W1-W2 interpretation and recount.

F4. W2 Time - Preserve the Contradictions

Source	W2 time guidance
Charting table	One section says W2 cannot take more time than W1; another graphic permits broader relationships.
Hardik	W1/4, W1/2, W1, or 2*W1.
Separate Time Analysis	W2 should be W1/4 or W1/2; if W2 time equals W1, it may actually be Wave B.
Impulse Notes	W2 time described as same as W1 or double W1.

SOURCE CONFLICT - DO NOT GUESS

This is a major conflict. Do not make time a hard invalidation until the owner selects a time preset. Recommended software behavior: calculate all source variants, show which pass, and keep W2 FORMING/TBD if time is the only unresolved item.

F5. HP BUY / HP SELL - Separate the Trade Signal from the Elliott Count

PROJECT-SPECIFIC REQUIREMENT

HP BUY/SELL is a project trading signal built on a valid W2 interpretation. It must never change the Elliott count just because a trade filter passes or fails.

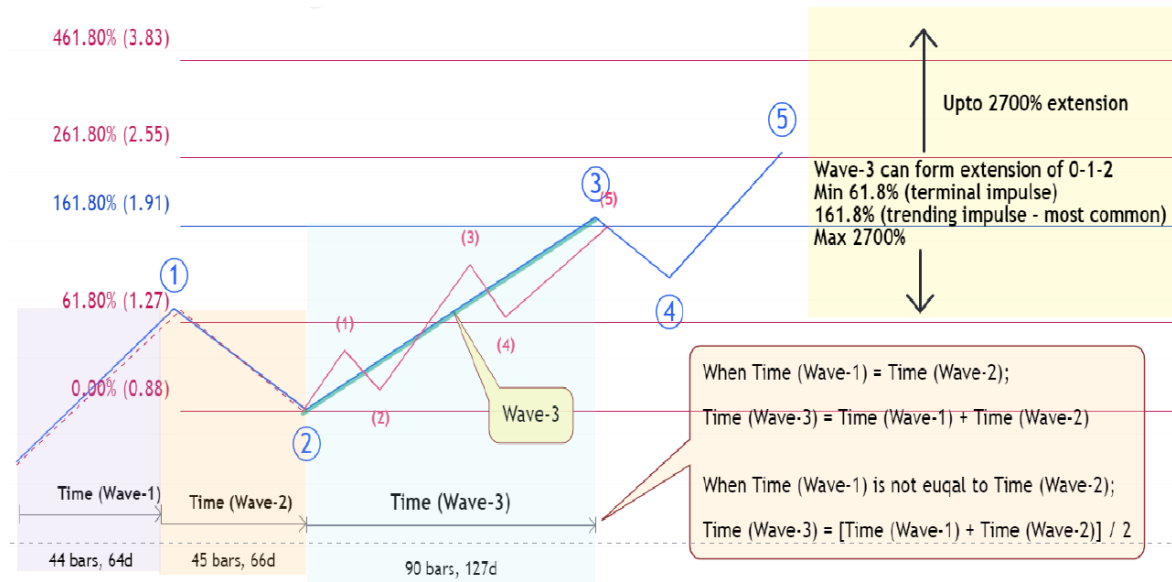
```

HP_BUY = W1_CONFIRMED
        AND w2_candidate.status in {FORMING_NEAR_COMPLETION, CONFIRMED}
        AND W2Retrace in 0.618..0.812
        AND Point0Valid
        AND correctionCompletionEvidence
        AND ownerSelectedTradeFilters

HP_SELL = mirror

```

G. Wave 3 - Trending, Terminal, and Extended

[Index](#)


Source visual: Wave-3 0-1-2 extension and time logic (Charting EW Notes p.11).

G1. W3 Must Never Be Determined Only by the Next High

Wave 3 needs an internal impulsive structure and a subtype classification. The source distinguishes a normal/trending degree Wave 3 (commonly at or beyond 161.8%) from terminal/subordinate behavior below 161.8%, and also allows extensions with 9/13/17/21 moves.

G2. W3 Projection

```
bullW3Ratio = (P3 - P2) / abs(P1 - P0)
bearW3Ratio = (P2 - P3) / abs(P0 - P1)

if ratio >= 1.618:
    TRENDING_W3 candidate
elif ratio >= terminalMin AND ratio < 1.618 AND terminalStructureValid:
    TERMINAL_W3 candidate
else:
    do not confirm degree W3
```

G3. W3 Is Not the Shortest

The “Wave 3 is never the shortest among 1,3,5” rule cannot be fully evaluated until Wave 5 exists. Therefore store it as a deferred hard rule. When W5 candidate forms, test W1, W3 and W5 lengths and reject the complete impulse if W3 becomes the shortest under the selected source interpretation.

G4. MACD and Gap Evidence

Evidence	Use
MACD top/highest	Support/rank the W3 candidate. Do not create W3 without price structure.
Breakaway gap above W1 high in bullish case	Source clue for W3; gap persistence rule is market/session dependent.
Price passes source extension but MACD does not show expected W3 behavior	Keep alternate correction/terminal interpretation alive rather than force W3.

G5. W3 Maximum Conflict

SOURCE CONFLICT - DO NOT GUESS

Chartking normalized specification states degree W3 can extend up to 2700%, while Impulse Notes show 2100%. Store the active maximum as a source-mode parameter. Never silently replace one with the other.

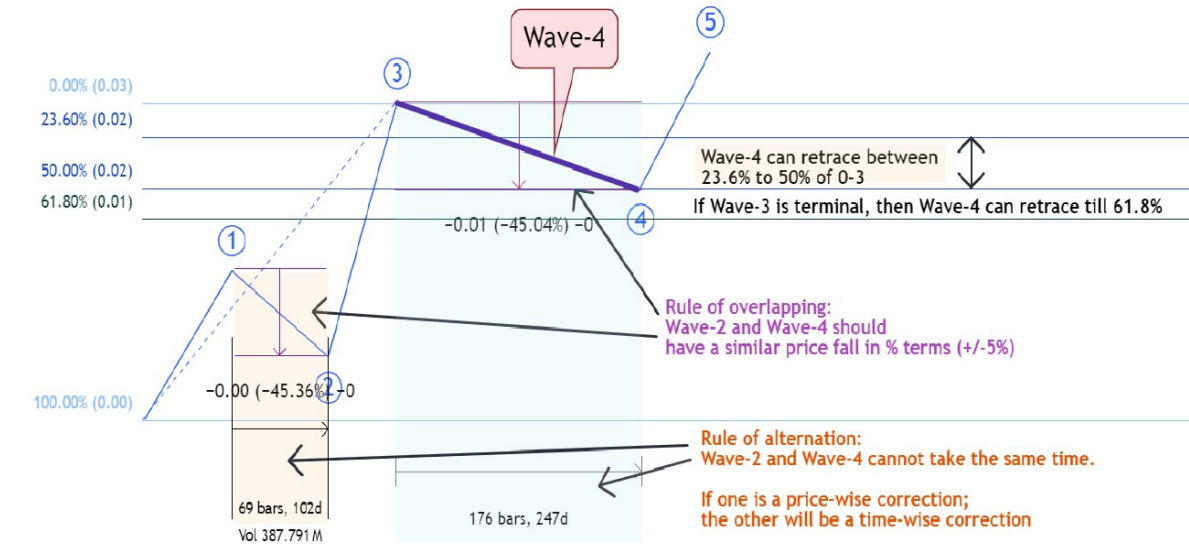
G6. W3 Time Conflict

Source	Time rule
Chartking	If $W1 \neq W2$, $W3 = (W1 + W2) / 2$; if approximately equal, $W3 = W1 + W2$.
Hardik	W3 minimum time described as $W1 + W2$.
Separate Time Analysis	W3 time = $W1 + W2$; also combines price equality / 127% W2 retracement confirmation.

SOURCE CONFLICT - DO NOT GUESS

Keep time-rule preset separate from price subtype. A valid trending W3 price structure should not be relabeled solely because an unresolved time source disagrees.

H. Wave 4 - The Main Source of Premature Wave 5 Errors

[Index](#)


Source visual: Wave-4 Base-to-W3 retracement, alternation and similarity (Chartking EW Notes p.13).

H1. W4 Is a Correction Container, Not One Leg

Wave 4 is where complex corrections are common. The engine should open multiple child correction candidates: Zig-Zag, Flat, Double, Triple and Triangle. Until one completes, the parent state is W4_FORMING. This is why the current forced A-B-C sequence is not enough.

H2. W4 Price Rules

Source mode	Normal W4 minimum	Normal max	Special/terminal
Chartking primary	23.6%	50%	May allow 61.8% in terminal context.
Hardik / Impulse Notes	14% appears as minimum in some tables	50%	Rare 61.8%.
W3 extension context	Context-specific	Often 38.2%, rare 50% in some notes	Complex correction expected.

SOURCE CONFLICT - DO NOT GUESS

As with W2, preserve a W4MinimumMode. Do not use one 14% constant everywhere.

H3. W4 Overlap Depends on Impulse Subtype

```

if impulseSubtype == TRENDING_NORMAL:
    apply normal overlap prohibition
elif impulseSubtype in {LEADING_DIAGONAL, ENDING_DIAGONAL, TERMINAL}:
    apply subtype-specific overlap rules
else:
    delay hard overlap decision until subtype can be evaluated

```

H4. Rule of Alternation and Similarity

The notes repeatedly use Wave 2 vs Wave 4 comparison. Price fall percentages may be similar within about +/-5% in the Chartking source, while time/structure should differ. This should be an explicit comparison object in code, not a visual-only note.

```
w2FallPct = abs(P1-P2) / referenceForW2
w4FallPct = abs(P3-P4) / referenceForW4
similarityDiff = abs(w2FallPct - w4FallPct)

priceSimilarityPass = similarityDiff <= ownerTolerance
timeAlternationPass = not timeApproximatelyEqual(w2time, w4time)
structureAlternation = type(W2) != type(W4) where source expects it
```

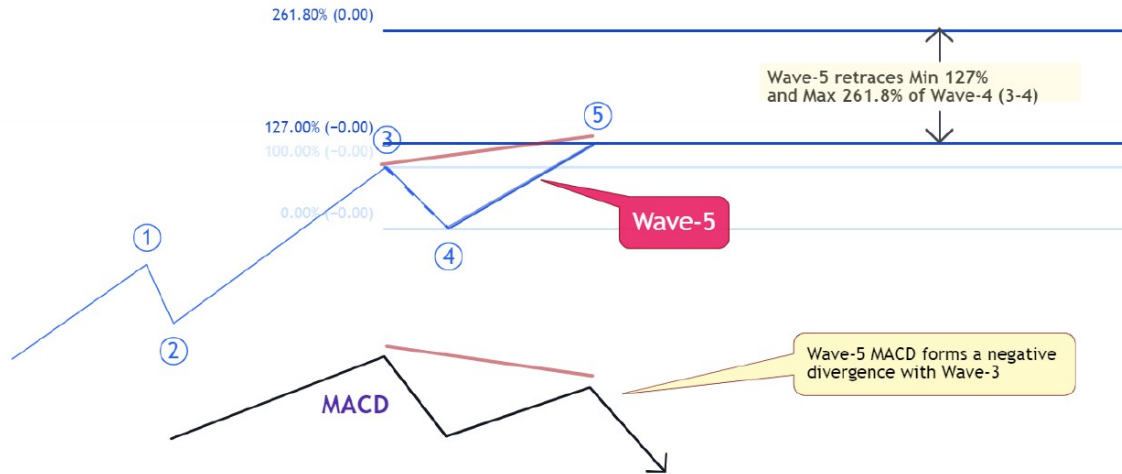
H5. W4 Time Conflicts

Source	W4 time options
Chartking summary	1/2, 2x or 3x W2; not same as W2.
Hardik	W2/2, 2x W2 or 5x W2.
Separate Time Analysis	W2/2; (W2+W3)/4; (W2+W3)/2; W2+W3.
W3 extension notes	W4 may be 5-10x / 10x W2 when W1/W2 microscopic.

SOURCE CONFLICT - DO NOT GUESS

Do not compress these into one arbitrary range. Calculate a list of expected windows per active source/subtype and show the nearest matching window in debug.

I. Wave 5 - Completion, Divergence, Truncation and Extension

[Index](#)


Source visual: Wave-5 127%-261.8% 3-4 projection and W3/W5 divergence (Charting EW Notes p.15).

11. Normal W5

```
normalW5PricePass = 1.27 <= W5Ratio <= 2.618
internalPass = fiveMoveImpulseCandidate
w3ShortestPass = not makesWave3Shortest(W1,W3,W5)
divergenceSupport = compareDegreeW3W5MACD()

if hard rules pass and required completion conditions pass:
    W5_CONFIRMED
```

12. Do Not Use One W5 Branch

Subtype	When to evaluate	Key distinction
Normal W5	Normal/trending impulse context	127%-261.8% 3-4 projection.
Truncated W5	Source double-extension context	Can terminate below normal 127 rule; source shows max around 81.2% of 3-4 while retaining five submoves.
W5 Extension / Megaphone	Specific source context, instrument restrictions/conflicts	Can extend toward source maximum; separate from normal W5.
Ending Diagonal W5	Terminal/late impulse context	3-3-3-3-3 and overlap/slow/divergence rules.

13. W5 Time Is Also Source-Dependent

Hardik says W5 usually takes W1 time and rarely W4 time. The separate Time Analysis document provides a wider set: W1, (W1+W3)/4, (W1+W3)/2, W1+W3 and an additional (W3+W4)/2 case. Keep these as selectable/diagnostic windows.

J. Trending vs Terminal Impulse, Leading Diagonal and Ending Diagonal

Leading Diagonal (5-3-5-3-5)

[Index](#)

Nature	• Terminal impulse	Feature-5	• Wave-5 is Min 161.8% and Max 261.8% , Wave-5 and Wave-1 may be equal
Sub-moves	• Sub-waves (1,3,5) should have 5 moves 1-2-3-4-5, all are slow	Time : W2	• Subordinate W2 = ¼ or ½ Subordinate W1
Occurrence	• Within Wave-1 and Wave-A (Zig-Zag correction)	Time:W3	• If W1 not equal to W2; then W3 = [W1 + W2]/2 • If W1 = W2 [+/- 5]; then W3 = W1 + W2
Retracement	• Wave-4 can retrace Min 23.2% and Max till 61.8% of 0-3	Next pattern	• Wave-2 (degree) after the entire leading diagonal should take ¼X, ½X or X
Feature-1	• Wave-4 may enter Wave-1's territory and will not go below end of Wave-2	Wave-3	• Wave-3 (degree) after Wave-1 is an LD, would mostly be extended
Feature-2	• Wave-3 Fib extension will be terminal Min 61.8% and Max <161.8%	Special feature	• Fall of Wave-2 (degree) after completion of the leading diagonal can go till sub-Wave-4, sub-Wave-2 or Max till 81.2%
Feature-4	• Wave-3 will be terminal and is slow in nature		

Source visual: Leading Diagonal rules and overlap behavior (Charting EW Notes p.18).

Ending Diagonal (3-3-3-3-3)

[Index](#)

Nature	• Terminal impulse	Feature-4	• Wave-3 is slow in speed
Sub-moves	• 3-3-3-3-3 (Triple Zig-Zag)	Feature-5	• Wave-5 is Min 127% and Max 261.8%, Wave-5 and Wave-1 may be equal in length
Occurrence	• Wave-5 and Wave-C (If Wave-3 is in extension, Wave-4 is a sharp move, then Wave-5 will be an ED)	MACD	• MACD forms negative divergence from Wave-3 (degree)
Retracement	• Wave-4 can retrace Min 23.2% and Max till 61.8% of 0-3	Time : W2	• Subordinate W2 = ¼ or ½ Subordinate W1
Feature-1	• Wave-4 enters Wave-1's territory and will not break Wave-2 low	Time:W3	• If W1 not equal to W2; then W3 = [W1 + W2]/2 • If W1 = W2 [+/- 5]; then W3 = W1 + W2
Feature-2	• Wave-3 Fib extension may be terminal Min 61.8% and Max <161.8%	Pattern End	• Correction after the entire ending diagonal should take 1/4 th or ½ its time
Feature-3	• Wave-2 and 4 won't retrace more than 50%	Criteria	• LD and ED will not form in the same impulse move (If W1 = LD, W5 = ED)

Source visual: Ending Diagonal rules and W5/C context (Charting EW Notes p.22).

J1. Trending vs Terminal Gate

Feature	Trending / normal	Terminal / diagonal family
W3 extension	Normally $\geq 161.8\%$ in the source main degree rule	May be $\geq 61.8\%$ but below 161.8% under terminal rules.
W4 overlap	Normal overlap prohibition	May enter W1 territory according to diagonal/terminal subtype rules.

Feature	Trending / normal	Terminal / diagonal family
W3 speed / channel	Usually stronger trending behavior	Often slower / repeated trendline interaction.
W5	Normal impulsive or extension	Can be diagonal depending on position/subtype.

J2. Leading Diagonal Parser

LD allowed only in source-approved positions (W1 and Wave A in the Chartking spec).

Expected high-level structure: 5-3-5-3-5.

Evaluate:

- W3 terminal range
- W4 overlap allowed but cannot violate LD-specific W2 boundary
- W4 retracement source range
- W5 source projection
- wedge / trendline behavior
- subordinate W2 time

Do not identify LD just because two trendlines converge.

J3. Ending Diagonal Parser

ED allowed only in W5 or Wave C.

Expected structure: 3-3-3-3-3.

Evaluate:

- valid position
- W2/W4 retracement limits
- overlap behavior
- W5 projection
- W3-vs-W5 MACD divergence
- post-pattern correction timing

ED_CONFIRMED only after all hard subtype conditions are met.

J4. LD/ED Coexistence Conflict

SOURCE CONFLICT - DO NOT GUESS

The source wording about whether Leading and Ending Diagonals can occur in the same impulse is internally ambiguous. Add a configuration or keep this criterion non-blocking/TBD until the owner selects the intended rule.

K. Extension Engine - 5, 9, 13, 17, 21 Moves

K1. Why “More Pivots” Does Not Mean “Next Main Wave”

An extended Wave 3 can contain 9, 13, 17 or 21 moves while the main degree is still Wave 3. This is exactly why a main count cannot be tied to consecutive raw pivots.

Move count	Meaning for engine
5	Normal internal impulse candidate.
9	First extension count in +4 sequence.
13	Further extension; still same parent wave.
17	Further extension; still same parent wave.
21	Maximum extension count described in source.

Source basis: *Chartking Developer Specification* p.11; *Hardik* pp.20-27; *Impulse Notes* pp.19-21.

K2. Generic Extension Rules

```
while parentWave.status == FORMING_EXTENDED:
    count internal confirmed moves
    require source minimum internal swing retracement/projection
    keep extension base fixed until degree-fall rule triggers
    if internalCount in {5,9,13,17,21}:
        evaluate completion candidate
    if degreeFallDetected:
        close parent extension / promote next-degree correction candidate
```

K3. Microscopic Extension

Hardik says microscopic W1/W2 may be hard to see, W1+W2 candle count can be small (e.g. ≤ 13 in that note), W3 then tends to extend, and W4 may stay shallow. Treat “microscopic” as a subtype with configurable quantitative thresholds because the source uses qualitative wording in other places.

K4. Megaphone / W5 Extension

Some supplied notes describe a W5 extension / megaphone condition when W1 and W3 are similar. The Developer Specification also carries instrument restrictions (source says not indices, only stocks) and equality wording conflicts. This must be its own subtype, not folded into normal W5.

SOURCE CONFLICT - DO NOT GUESS

W5 extension applicability and W1/W3 equality criteria vary by source. Keep instrument gating and equality tolerance configurable.

L. Correction Classifier - The Most Important Refactor After P0

L1. Parallel Candidate Model

```
onCorrectionStart(parentWave):
    candidates = []
    if positionAllowsZigZag: candidates += ZIGZAG
    if positionAllowsFlat:   candidates += FLAT
    if positionAllowsDouble: candidates += DOUBLE
    if positionAllowsTriple: candidates += TRIPLE
    if positionAllowsTriangle: candidates += TRIANGLE

onSubPivot(p):
    for c in candidates:
        c.evaluate(p)
        if c.hardInvalid: remove(c)

    rankRemainingCandidates()
    if one candidate fully confirmed: promote it
    else show PRIMARY + ALTERNATE in debug
```

IMPLEMENTATION WARNING

Do not ask the user to choose "A-B-C / W-X-Y / W-X-Y-X-Z / A-B-C-D-E" so the script knows what to label. That dropdown can remain only as a display filter or manual debug override.

M. Zig-Zag - Exact Coding Checklist

6. Correction classifier - simple correction

Correction recognition must be candidate-based because the source explicitly calls corrections guideline based. The same early A-B path may later resolve to Zig-Zag, Flat, Double, Triple or Triangle.

6.1 Zig-Zag Correction (A-B-C)

Field	Source requirement
Type / occurrence	Single, price-wise correction; mostly Wave-2, sometimes Wave-4 and inside Double/Triple.
Count	3 moves A-B-C; internal 5-3-5.
Speed	A fast, B slow, C fast.
Wave-A	Min 61.8% retracement of previous degree Wave-5.
Wave-B time	$B \geq A$, up to 5X A.
Wave-B price	Less than 61.8%; source says if >61.8% it is Wave-2.
Wave-C projection	$C = A$ length-wise; min 61.8%, source table max 261.8% extension of 0-A-B.
Wave-C retracement	Min 127% of Wave-B.
C time	If $A=B$, $C=A+B$; if $A < B$, $C=(A+B)/2$.
Entry	On crossing the 3rd Fib ratio in Wave-C.
SL	Previous Fib level of entry.
Targets / reversal support	Bullish candle at 61.8%, 100%, 161.8%, 261.8%; 0-B trendline break in half C time; point B crossed in same C time.
Pattern end	Complete correction minimum 38.2% of entire impulse.
C types	Normal 61.8%-161.8%; Elongated source also shows up to 461.8%; Truncated reverses before 61.8% and uses an 81.2% confirmation.

Zig-Zag Correction (A-B-C)

[Index](#)

Correction Type	•Single correction	Entry	• On crossing 3 rd Fib Ratio in Wave-C	
Occurrence	•Mostly found in Wave-2/ sometimes in Wave-4 and in double or triple correction	SL	• Previous Fib Level of entry	
Category	•Price-wise correction	Target	• Bullish Candle formation at Fib levels 61.8%, 100%, 161.8%, 261.8%	
No. of moves	•3 (A-B-C)	Pattern confirmation-1	• When price crosses 0-B trendline in half the time (Wave-C)	
Internal count	•5-3-5	Pattern confirmation-2	• When price crosses point B at the same time (Wave-C)	
Speed	•A Fast, B Slow, C Fast	Pattern End	• Complete correction should retrace Min 38.2% of the entire Impulse move	
Rule for Wave-A	•Min 61.8% retracement of previous degree Wave-5	Types of C : Trend based Fib extension on 0-A-B	Normal C	61.8% to 161.8%
Wave-B Time period	• B ≥ A (up to 5 times of A)		Elongated C	161.8% to 461.8%
Fib Ext Wave-C (0-A-B)	• C = A (length wise); Min 61.8%; Max 261.8% Fib extension (0-A-B)		Truncated C	1. Reverses before 61.8%
Wave-C retracement	• Wave-C should be Fib retracement 127% of Wave-B			2. Confirmation: 81.2% retracement of A-B-C (0-C)
Wave-B retracement	• Less than 61.8%, if >61.8% then it is Wave-2			
Time taken for Wave-C : 1	• If A = B, C = A + B			
Time taken for Wave-C : 2	• If A < B, C = (A + B) / 2			

Source snapshot - page 37: complete Zig-Zag rule table.

Rule	Code check	State effect
Structure	A-B-C; internal 5-3-5.	Hard structural requirement for confirmed Zig-Zag.
A	Source min 61.8% of previous-degree W5 in this pattern context.	Below threshold -> remain FORMING/alternate according to source.
B price	Below Flat-style 61.8%; detailed summary gives 1%-50%.	If B enters Flat range, Flat candidate becomes stronger.
B time	B >= A; source up to 5A for Zig-Zag.	Time support / conflict mode.
C projection	0-A-B extension; normal/elongated/truncated subtypes.	Determines C subtype and target zone.
C vs B	Source min 127% retracement of B.	Required where active source says mandatory.
Trendline	0-B break in source timing.	Completion/reversal support.
Whole correction	Min 38.2% of entire impulse in source table.	Pattern-end check.

N. Flat - Exact Coding Checklist

Flat Correction (A-B-C)

[Index](#)

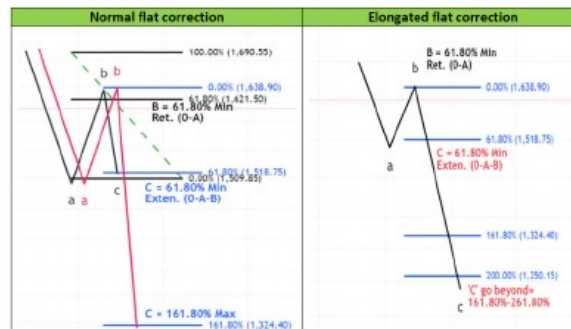
Correction Type	• Single correction	Point of Entry	• Start of Wave-X or Wave-1	
Occurrence	• Mostly found in Wave-2	Confirmation:1	• Bullish Candle formation at Fib levels 61.8%, 100%, 161.8%, 261.8%	
Category	• Price-wise correction	Confirmation:2	• On crossing 3 rd Fib Ratio	
No. of moves	• 3 (A-B-C)	SL	• 61.8%, 100%, 161.8%, 261.8%	
Internal count	• 3-3-5	Exit (time period)	• Half the time of Wave-C	
Speed	• A Slow, B Fast, C Fast	Reversal criteria	• When price crosses 0-B trendline in half the time (Wave-C)	
Rule for Wave-A	• Min 38.2% retracement of previous degree Wave-5	Pattern confirmation	• When price crosses point B at the same time (Wave-C)	
Rule for Wave-B	• B >= A (up to 2 times of A)	Pattern End	• Complete correction should retrace Min 38.2% of the entire Impulse move	
Wave-B retracement	• Min 61.8% , Max 111%, (SL 127%)			
Time taken for Wave-C : 1	• If time (A) = time (B), time (C) = time (A) + time (B)			
Time taken for Wave-C : 2	• If time (A) < time (B), time (C) = [time (A) + time (B)]/2			

Types of C : Trend based Fib extension on 0-A-B	Normal C (B : Min 61.8%, Max 81.2%)	61.8% to 161.8%
	Elongated C (B: Min 61.8%, Max 81.2%)	161.8% to 261.8%
	Strong Irregular (B: Min 61.8%, Max 111%)	61.8% to 161.8%
	Running (B: Min 61.8%, Max 111%)	161.8% to 261.8%

Source snapshot - page 43: Flat correction rule table.

[Index](#)

Normal flat correction	Elongated flat correction	Truncated flat correction	Irregular flat correction (Strong-B)	Running flat correction
A= max61.80%	A= max61.80%	A= max61.80%	A= max61.80%	A= max61.80%
B=61.80% min = 81.20% max	B=61.80% min = 81.20% max	B=61.80% min = 111% max	B=Between 100% to 111%	B=Between 100% to 111%
C=61.20% min 161.80% max	C=Go beyond 161.80%-261.80%	C=does not go beyond 61.80%	C=61.20% min 161.80% max	C=above 161.80% -261.80%



Source snapshot - page 47: Normal, Elongated, Truncated, Irregular and Running Flat comparison.

```

Flat candidate gates:
  structure = 3-3-5
  A >= selected source minimum (38.2% in Chartking Flat table)
  B retracement in 61.8%-111%
  classify C subtype from B range + C projection
  evaluate time and 0-B reversal rules

```

Subtype	B	C	Developer note
Normal Flat	61.8%-81.2%	61.8%-161.8%	Normal C candidate.
Elongated Flat	61.8%-81.2%	161.8%-261.8% in one table	Do not silently extend to 461.8 without owner rule.
Strong Irregular	61.8%-111%	61.8%-161.8%	B can exceed A while remaining below source max.
Running Flat	61.8%-111%	161.8%-261.8% in summary	Preserve structural requirements, not only ratios.

O. Wave B - How to Prevent the Old Premature Wave 5 Bug

The old bug happened because the engine expected a fixed sequence. The correct fix is not merely “allow B to 111%”. The correct fix is to keep correction candidates alive until their structure resolves.

```

After a candidate A:
  if B < 0.618 of A:
    keep ZigZag candidate (if internal rules allow)
  if 0.618 <= B <= 1.11:
    keep Flat candidate
  if correction is already complex:
    keep Double/Triple/Triangle candidates according to position

```

Only after W4 correction completion can the next parent impulse leg be evaluated as W5.
Never infer W5 from “next pivot number”.

P. Wave C - Completion Engine

Wave C is not simply the third pivot in A-B-C. It is an impulsive child wave (or ED where allowed), has source-specific Fib/time/MACD behavior, and its completion is important for both correction confirmation and project signals.

```
evaluateC(corr):
    build C internal impulse / ED candidate
    cExt = extension012(originA, A, B, currentC)
    cVsB = abs(C-B) / abs(B-A)
    cTime = duration(B,C)

    evaluate normal / elongated / truncated subtype
    evaluate positive divergence support
    evaluate 0-B trendline / point-B timing support

    if completion conditions pass:
        corr.C = CONFIRMED at actual completion pivot
    else:
        keep C FORMING
```

PROJECT-SPECIFIC REQUIREMENT

C (and later Z) should be marked on the actual completion pivot/candle when the pattern completes. Do not shift the label to a later unrelated swing just to make the count visually neat.

Q. Double Correction W-X-Y

8. Double Correction (W-X-Y)

Field	Source requirement
Type / occurrence	Complex, time-wise correction; mostly Wave-4.
Internal structure	W and X: Zig-Zag or Flat; Y: Zig-Zag, Flat or Triangle.
Move count	Source summary gives max 39 with W(5-3-5), X up to 11, Y(3-3-3-3-3); later examples provide specific totals by combination.
Wave-X small	Max 50% retracement of W.
Wave-X large	Min 61.8%, max 111% retracement of W.
Wave-Y fib	0-W-X extension: min 61.8%, max 161.8%.
Wave-Y time	$\text{Time}(Y) = \text{Time}(W)$ OR $\text{Time}(W+X)$ OR $\text{Time}(W+X)/2$.
Wave-X time	$\text{Time}(X) < \text{Time}(W)$.
Entry evidence	Positive MACD divergence at Y.
Confirmation	Source cell is literally "???" - must remain TBD.
Entry	Close above 38.2% retracement of 0-Y.
SL	Trailing SL of 2 Fib levels; if price crosses 81.2%, source says SL at 50%.
Reversal	0-X TL crossed in half $\text{Time}(Y)$; Point B crossed in less than $\text{Time}(Y)$.

Double Correction (W-X-Y)

[Index](#)

Correction Type	• Complex correction	Time taken for Wave-Y	• $\text{Time}(Y) = \text{Time}(W)$ OR $\text{Time}(W+X)$ OR $\text{Time}(W+X)/2$
Occurrence	• Mostly found in Wave-4	Time taken for Wave-X	• $\text{Time}(X) < \text{Time}(W)$
Category	• Time-wise correction	Point of Entry	• Positive divergence in MACD at Y
No. of moves	• Total Max 39 -> W(5-3-5), X(11), Y(3-3-3-3-3)	Confirmation	• ???
Internal structure	• W & X: Zig-Zag or Flat; Y: Zig-Zag, Flat or Triangle	Entry	• Close above 38.2% retracement of 0-Y
Rule for Wave-W	• If 1 st Wave takes 1/2, 1/3 or 1/4 Time of 2 nd Wave, then 1 st Wave is "W" and 2 nd Wave is "X"	SL	• Trailing SL of 2 Fib levels (SL will be at 50% if price crosses 81.2% Fib level)
Rule for Wave-X (small)	• Max 50% retracement of Wave-W	Exit (time period)	
Rule for Wave-X (Large)	• Min 61.8% and Max 111% retracement of Wave-W	Reversal criteria:1	• When price crosses 0-X TL in 1/2 Time (Wave-Y)
Rule for Wave-Y	• 0-W-X Extension: Min 61.8%, Max 161.8%	Reversal criteria:2	• When price crosses Pt. B in < Time (Wave-Y)

Source snapshot - page 56: Double Correction rule table including unresolved confirmation field.

Q1. Structure

W = simple correction candidate (ZigZag or Flat)
 X = connector correction
 Y = second correction candidate (ZigZag, Flat or Triangle)

Do not represent W, X and Y as three arbitrary raw pivots. Each component is itself a structure/container.

Q2. X and Y Rules

Component	Source rule	Coding action
Small X	Max 50% of W.	Classify X_SMALL.
Large X	61.8%-111% of W.	Classify X_LARGE.
X time	$X < W$.	Store pass/fail and selected tolerance.
Y Fib	0-W-X extension 61.8%-161.8%.	Project from X using W length.
Y time	W, W+X, or $(W+X)/2$.	Generate expected windows, not one continuous max.
Y MACD	Positive divergence at end in source.	Supporting completion evidence.
Confirmation	Source cell contains "???".	TBD-BLOCKED; do not invent textbook confirmation.

R. Triple Correction W-X-Y-XX-Z

10. Triple Correction (W-X-Y-XX-Z)

Field	Source requirement
Type / occurrence	Complex, time-wise; mostly Wave-4.
Initial structure	W & X: Zig-Zag or Flat; Y: Zig-Zag, Flat or Triangle.
Small X	Max 50% of W (another page: min 14.2%-max50%).
Large X	61.8%-111% of W.
Y	0-W-X extension min61.8%-max161.8%.
XX	50%-61.8% retracement of Y; cannot go above Point X.
Z	61.8%-161.8% extension of 0-Y-XX; source says failure may reverse from 31.8%.
Time X	$X < W$; another diagram says $X < 1/2 W$.
Time Y	$Y = W$ OR $W+X$ OR $(W+X)/2$.
Time XX	$XX = X$ and $< Y$.
Time Z	$Z = Y$ OR $Y+XX$ OR $(Y+XX)/2$.
Point of Entry	Start of Wave-Y in the source summary.
Blank source fields	Rule for W, Rule for XX/Z in summary, Confirmation, Entry, SL and Exit cells are blank. Use detailed later pages only where they explicitly provide values.

Triple Correction (W-X-Y-XX-Z)

[Index](#)

Correction Type	• Complex correction	Time taken for Wave-Y	•Time (Y) = Time (W) OR Time (W+X) OR Time (W+X)/2
Occurrence	• Mostly found in Wave-4	Time taken for Wave-X	•Time (X) < Time (W)
Category	• Time-wise correction	Point of Entry	•Start of Wave-Y
No. of moves	• Total Max 37 -> W(5-3-5), X(11), Y(5-3-5)	Confirmation	
Internal structure	• W & X: Zig-Zag or Flat ; Y: Zig-Zag, Flat or Triangle	Entry	
Rule for Wave-W		SL	
Rule for Wave-X (small)	• Max 50% retracement of Wave-W	Exit (time period)	
Rule for Wave-X (Large)	• Min 61.8% and Max 111% retracement of Wave-W	Reversal criteria:1	•When price crosses 0-X TL in 1/2 Time (Wave-Y)
Rule for Wave-Y	• 0-W-X Extension: Min 61.8%, Max 161.8%	Reversal criteria:2	•When price crosses Pt. B in < Time (Wave-Y)
Rule for Wave-XX			
Rule for Wave-Z			

Source snapshot - page 73: Triple Correction summary with blank fields.

R1. Exact Component Model

TRIPLE = W + X + Y + XX + Z

W = first correction
 X = first connector
 Y = second correction
 XX = second connector (do NOT call it another generic X internally)
 Z = final correction

Component	Source rule / range
Small X	Max 50% of W; another page gives min14.2%-max50%.
Large X	61.8%-111% of W.
Y	0-W-X extension 61.8%-161.8%.
XX	50%-61.8% retracement of Y; cannot go above Point X.
Z	0-Y-XX extension 61.8%-161.8%; failure percentage has source conflict.
Time X	$X < W$; another diagram says $X < 1/2 W$.
Time Y	$Y = W$ OR $W + X$ OR $(W + X)/2$.
Time XX	$XX = X$ and $< Y$.
Time Z	$Z = Y$ OR $Y + XX$ OR $(Y + XX)/2$.

IMPLEMENTATION WARNING

The current display option "W-X-Y-X-Z" should be changed internally to W-X-Y-XX-Z so the two connectors have distinct identities, histories and time/Fib rules.

S. Triangle Engine - Six Families, Not One ABCDE Label

11. Triangle Correction engine

11.1 Triangle families and common rules

Family	Types
Contracting	Horizontal Contracting, Irregular Contracting, Running Contracting.
Expanding	Horizontal Expanding, Irregular Expanding, Running Expanding.

Common contracting rule	Source requirement
Retracement condition	Three waves must retrace more than 50% of the preceding wave length.
Structure	A-B-C-D-E; each wave is corrective.
Trendlines	Always draw on candle chart; both trendlines max 5 touch points.
Wave-E guideline	Typically around 38.2% retracement.
Where triangles appear	Mostly Wave-4, B, E, X, Y, XX, Z.
Exclusion	Not found in Wave-2.

Types of Triangles

[Index](#)

Contracting Triangle:

- Horizontal Contracting Triangle.
- Irregular Contracting Triangle.
- Running Contracting Triangle

Expanding Triangle:

- Horizontal Expanding Triangle.
- Irregular Expanding Triangle.
- Running Expanding Triangle.

Source snapshot - page 87: six triangle types.

S1. Common Rules

Rule	Developer interpretation
Structure	A-B-C-D-E; every leg is corrective.
Location	Mostly W4, B, E, X, Y, XX, Z in the source; not W2.
Trendlines	Triangle boundaries must be built from the correct leg pivots; source limits touch-point behavior.
Retracement relation	Common contracting rule requires multiple legs to retrace >50% of preceding leg.
E guideline	Often near a source Fib guideline; exact subtype rules differ.
Thrust	Calculate only after confirmed triangle completion; subtype determines expected thrust.

S2. Horizontal Contracting

```
size(A) > size(B) > size(C) > size(D) > size(E)
trendlines converge / move in opposite price directions
A = widest
source thrust ~75%-125% of widest wave
apex relation must be evaluated where source mode requires it
```

S3. Irregular Contracting

The source permits B to be the widest segment, often below 161.8% of A and not above the larger maximum described in the notes. C, D and E then contract sequentially. The thrust is again tied to the widest wave in the common rule.

S4. Running Contracting

Running Contracting has different trendline slopes and size ordering. The supplied source also contains conflicting thrust statements (normal 75%-125% vs exceptional much larger thrust). Preserve this as a source conflict.

S5. Expanding Family

Expanding triangles reverse the size concept: A or B is among the smallest, E is usually the largest, and E often breaks the A-C boundary. The Hardik notes also give a 61.8% of E thrust guideline for expanding triangles. Do not apply the contracting thrust formula to expanding patterns automatically.

Source basis: Developer Specification pp.24-27; Hardik triangle section; Charting triangle pages.

T. MACD Engine - Support the Structure, Never Replace It

Stage	Source clue	How code should use it
Base / W1	MACD lowest low or positive divergence.	Base score / start evidence.
W2	MACD falls slowly; mostly positive divergence; selected source may protect base MACD.	Support W2, not a standalone label.
W3	MACD highest / forms top.	Support trending W3.
W4	MACD may fall rapidly / become very low in price-wise correction.	Degree-fall / W4 support.
W5	MACD lower than W3; negative divergence.	Completion support / ED clue.
A	MACD lower than prior impulse W2/W4 in source correction rules.	Correction classification support.
B / X	MACD moves with speed.	Connector/B support.
C	Positive divergence can indicate simple correction completion.	C completion support.
Y	Positive divergence; lowest-low behavior may signal triple continuation in source.	Double vs Triple clue.
XX	MACD higher high.	Triple connector support.
Z	Positive divergence.	Triple completion support.

Source basis: *Charting MACD Rules - Impulse/Correction; Developer Specification pp.28-30.*

SOURCE CONFLICT - DO NOT GUESS

The source does not specify exact MACD fast/slow/signal parameters or exact divergence pivot algorithm. Keep these inputs configurable. The current code should also expose whether it is comparing MACD line, histogram, or both.

U. Gaps - Supporting Context Only

The supplied Impulse Notes and Developer Specification place different gap classes in specific wave contexts. Treat gaps as supporting evidence and be careful in 24-hour markets where “opening gap” behavior differs.

Gap	Source contexts	Implementation
Common	W1/W2/W4, Flat/Triangle/B/X contexts in supplied notes.	Small support tag; track fill window if applicable to instrument.
Breakaway	Primarily W3 after W1 breakout / news.	Support strong W3, do not force W3.
Runaway	W3, sometimes Zig-Zag A / X in supplied notes.	Trend-strength support.
Exhaustion	W5, C, Y, Z near maturity.	Completion/reversal support.

V. Trendline and Channel Engine

TRENDLINES

- There are three types of trendline namely:-
 1. BASE TRENDLINE-Drawn between 0 point to 2 point and extended.
 2. ACTION TRENDLINE-Base trendline is copy paste and then put to wave 1 is called action line.
 3. MATURITY TRENDLINE-This trendline is drawn between wave 2 and wave 4.

Hardik source page showing Base, Action and Maturity trendline definitions (p.39).

14. Trendline and channel engine

14.1 Impulse channel

Rule	Source requirement
Activation	When price crosses 61.8% of Important High-Low, draw channel to check W1 trend continuity.
Use	Impulse only.
Construction	Join base point with 2; ensure price does not fall outside the channel.
Exit meaning	After channel exit, price can fall 38.2%-81.2%, signifying end of trend.
Re-entry note	Entry after fall with reversal candle; source says price does not re-enter channel after exit.

Channel

[Index](#)

Whenever price crosses 61.8% of the Imp High – Low, draw a channel to check trend continuity of Wave-1

To be used only in Impulse waves. Price won't reverse till the time it does not exit the channel

Make it by joining the base point with 2 and make sure that price does not fall anywhere outside the channel

After exiting from the channel, price can fall till 38.2% to 81.2% levels and signifies end of trend

Entry can be done after the fall with a reversal candle pattern. Price does not enter the channel after exiting it

Source snapshot - page 108: Channel rules.

14.2 Trendline construction and projections

Stage	Source requirement
Base channel	Connect 0-2 and parallel through 1. No part of Wave-3 should break 0-2 TL.
Wave-4 projection	Connect 1-3 and parallel through 2; W4 can occur where line through 2 matches Fib levels.
Maturity line	While price remains above 2-4 TL, source expects Wave-5 to reach 127%.
Wave-C projection	Connect origin of A with B and place parallel line through A to project end of C.
After W3	Connect 1-3 and parallel through 2 to estimate W4 boundary.
After W4	Connect 2-4 and parallel through 3 to forecast W5 end.
Throw-over	If W4 moves out of channel, expect W5 throw-over.
Alternative W5 method	If W3 very strong/vertical and W4 sideways, connect 2-4 and parallel through W1 extreme.

V1. Objects to Store

```
TrendlineRecord {
  type: BASE_0_2 | ACTION_PARALLEL_1 | MATURITY_2_4 | CORRECTION_0_B | TRIANGLE_BOUNDARY,
  anchor1Bar, anchor1Price, anchor2Bar, anchor2Price,
  slope, projectedPrice, breakBar, breakPrice, breakConfirmed, sourceRuleId
}
```

V2. Impulse Lines

Line	Construction	Use
Base / 0-2	Connect 0 to 2.	Trend continuity / W3 weakness clue.
Action	Parallel copy through W1.	Strong W3 behavior when price travels above/near action line in source.
1-3 / projection	Connect 1-3 and parallel through 2.	W4 projection/channel boundary.
Maturity / 2-4	Connect 2-4.	W5 projection and post-W5 reversal confirmation.
0-B correction line	Connect correction origin to B.	ABC completion/reversal timing support.

V3. Reversal Timing after W5

Hardik material says to measure the time of the 4-to-5 leg; a maturity-line break within about half that time is a reversal confirmation clue, while reaching Wave 4 within an equal time is a pattern confirmation clue. Keep these as source-specific post-W5 confirmation rules, not universal Elliott textbook assumptions.

W. Time Analysis - Keep It as a Separate Engine

Time Analysis

Confidence in trading can be achieved using study of market structure (Elliott Wave theory + Fib ratios) + Time analysis + Price Action (Candlestick patterns, indicators)

Market remains in correction 80% of the time and in impulse only 20% of the time.

Time period is a different concept than Elliott waves. While Elliott wave (base point in case of flat correction) may begin from a lower high than the price top (where top is 111%), time period is to be considered from market high and lows only (not to be misunderstood/mixed with base point 0 in case of flat correction)

Time analysis

1. Elliot Wave time periods (tp)
2. Fib Wave proportions (wp)
3. Fib sequence Numbers (fn)

Few important points about time period

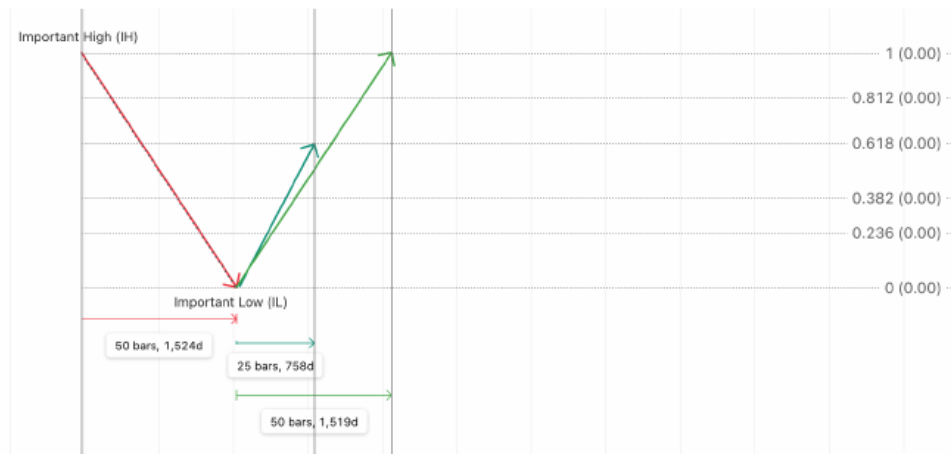
- a. Reversal zones: Points in time where all these time periods come together to form clusters and act as probable reversal zones
- b. High/Low price levels of 1/4th time period are most important for trading purposes

Elliott Wave Time Periods

Wave 1 time period: Wave 1 often starts as impulse after a correction.

What qualifies a good impulse:

1. When price recovers at least 61.8% of previous move (usually, downfall in stock price) from Important high (IH) to Important low (IL) in about $\frac{1}{2}$ of the period taken by previous move (IH to IL)
2. When price recovers 100% of the fall in about the same time (time from IH to IL)



Impulse if **confirmed** only after formation of wave 3 (sharp upmove). Before that it is just an assumption/possibility. Fast up moves after sharp fall (correction) can also be X wave, after which the price will most likely enter into double / triple correction.

Separate Time Analysis source: time origin and Wave-1 concept (p.1).

Wave 2 time period (W2 tp):

1. Wave 2 time period (tp) can never be equal to wave 1 time period (tp).
2. It will be either $\frac{1}{2}$ or $\frac{1}{4}$ of the wave 1 time period
3. If wave 2 tp is equal to wave 1 tp, it may not be wave 2 but wave B which (tp) is equal to wave A
4. Wave 2 is sharp most of the time as it is zig zag correction (price correction more often than time correction). This happens as market is generally not very bullish at this point after fall from important high to important low (correction)
5. Wave 2 can retrace between 23.6% to 81.2% in about $\frac{1}{2}$ or $\frac{1}{4}$ of the wave 1 time period.

Summary:

- a. $\frac{1}{2}$ of wave 1 time period
- b. $\frac{1}{4}$ of wave 1 time period

Wave 3 time period (W3 tp):

1. Wave 3 formation can only be confirmed by two ways
 - a. Following rule of equality where wave 3 is equal to price range of wave 1. This can be checked by applying three point fib extension tool from base point 0 to high/end of wave 1 and low/end of wave 2. It should be 100% i.e. same price range as wave 1
 - b. As retracement of wave 2 price, being minimum 127%. This can be verified using fib retracement from high/end of wave 1 to low/end of wave 2 towards wave 3.
2. Most of the time fib extension of 100% and 127 retracement will be at similar level creating confluence of price level.
3. ## Wave 3 is generally sharp move supported by price action indicators such as upper bollinger band (std. 2) turning upwards, RSI>60 and ADX above 12-15 and rising
4. Time period for wave 3
Wave 1 time period + Wave 2 time period : (W1 tp + W2tp)

Wave 4 time period (W4 tp)

1. Wave 4 is a corrective move after wave 3
2. As per theory of alternation, time period of wave 2 and wave 4 will not be the same, through correction in terms of price range % (rule of similarity) will be similar. This is because if wave 2 is simple zig zag price correction, more often wave 4 is time correction forming all type of patterns, zig-zag, flat, triangle, double and triple correction.
3. Time period for wave 4
 - a. Wave 2 time period divided by 2 : $W2\ tp / 2$
 - b. Wave 2 time period plus Wave 3 time period, divided by 4 : $(W2\ tp + W3\ tp) / 4$
 - c. Wave 2 time period plus Wave 3 time period, divided by 2 : $(W2\ tp + W3\ tp) / 2$
 - d. Wave 2 time period plus Wave 3 time period : $(W2\ tp + W3\ tp)$
4. If wave 4 is taking more time than W2 and W3 combined, it may be something else or the wave count is incorrect

Separate Time Analysis source: Wave-2 to Wave-4 time rules (p.2).

W1. Critical Concept: Time Origin May Differ from Elliott Base

The separate Time Analysis document explicitly warns that time should be measured from market highs/lows and not confused with an Elliott base point in a Flat correction. Therefore store Elliott price anchors and time-analysis anchors separately.

```
WaveCandidate.priceOriginBar // Elliott structural origin
WaveCandidate.timeOriginBar  // market high/low origin if selected time-source requires it

Never assume priceOriginBar == timeOriginBar in every correction.
```

W2. Time Windows, Not One Scalar

Because the sources provide multiple discrete expected times, represent time expectations as a list of windows rather than a single maxTime number.

```
expectedWindows = [
  {name:"W1/4", expected: W1time*0.25},
  {name:"W1/2", expected: W1time*0.50},
  ... active source-specific options ...
]

nearest = min(relativeDistance(actualTime, each.expected))
timePass = nearest.distance <= selectedTolerance
```

W3. Fib Time Zones and Cluster Zones

The separate Time Analysis notes introduce Fib sequence time zones (13, 21, 34, 55, 89, 144, 233, 377, 610, 987) and cluster zones where wave time periods, wave proportions and Fib time zones converge. This is an additional time-analysis module, not a replacement for price-structure Elliott rules.

Source basis: *Time Analysis pp.5-7.*

W4. Trade-Timing Module Is Separate from Wave Identification

Trade setup for Waves

Trading for Wave 3

1. Wave 1 and Wave 2 are not preferred for trading
2. When wave 2 is in development or completed, mark high of $\frac{1}{4}$ and high of $\frac{1}{2}$ time period candles ($\frac{1}{4}$ and $\frac{1}{2}$ of wave 1 time period marked from start of wave 2 <end of wave 1>)
3. When price starts moving up for wave 3 (preferably after wave 2 time period is complete and price has made a bottom <38% to 81.2% retracement of wave 1>), enter above the high of $\frac{1}{4}$ th time period candle
4. If entry was made at $\frac{1}{2}$ time period candle, first target would be high of $\frac{1}{4}$ th time period candle

Target achievement / Exit

1. Exit when earlier of the below happens
 - a. Wave 3 has reached 100% of wave 1 (fib extension), rule of equality has been achieved
 - b. 127% retracement of wave 2 has reached
 - c. Wave 3 time period is over ($W1\ tp + W2\ tp$)

Time period should be the first priority as it prevents us from taking trade in haste.

Trading for Wave 5

Entry

1. Priority 1: Apply rule of similarity to validate if price range (% fall) of wave 4 is same as wave 2 (+/- 5%, [10% in case of CMP < 500])
2. Priority 2: Check for formation of bullish candle pattern at important fib retracement level (38% or 50% of entire upmove from base point zero)
3. Priority 3: Time period of wave 4 is complete

Enter where confluence of more than one trigger is occurring (where price fall is similar to wave 2 and bullish candle pattern is forming at important fib level, which is also acting as an important price support level).

Exit

1. Exit when price touches 127% retracement level of wave 4 or time period of wave 5 is over ($(wave\ 1\ time\ period + wave\ 3\ time\ period)/4$, whichever is earlier).

Time Analysis trade examples: W3/W5 trade timing and target logic (p.4).

If implemented, trading entries/exits must consume Elliott state; they must not change the Elliott labels. For example, a W3 trade can wait for time/Fib triggers even while W3 is structurally FORMING. Likewise, a failed trade filter does not make the Elliott count invalid.

X. Source Conflict Matrix - Must Be Visible to the Developer

Topic	Chartking / normalized spec	Hardik / Impulse / Time notes	Required software treatment
W2 minimum	Normal 23.6%; 14% microscopic special	14% or 14%-23.6% shown as minimum in other notes	Preset / subtype; no silent merge.
W2 time	Several versions; one says $\leq W1$	Hardik 1/4,1/2,1,2; Time Analysis 1/4 or 1/2; Impulse notes same/double	Discrete source presets.
W3 time	Average if $W1 \neq W2$, sum if equal	Hardik/Time Analysis use $W1+W2$ as primary	Preset.
W3 maximum	2700% in normalized spec	2100% in Impulse Notes	Configurable source max.
W4 minimum	23.6% normal	14% in Hardik/Impulse tables	Preset/context subtype.
W4 time	1/2,2x,3x W2	Hardik 1/2,2x,5x; Time Analysis uses $W2+W3$ formulas	List of source windows.
W4 extension max	38.2 vs 50 in extension notes	Rare 50 etc.	Context + conflict mode.
W5 time	Not one universal rule	Hardik $W1$ /rare $W4$; Time Analysis has multiple $W1/W3$ formulas	Separate time preset.
C elongated max	261.8 in some tables	Up to 461.8 in other diagrams/notes	Conflict mode.
Double confirmation	Source cell "???"	No unambiguous replacement	TBD-BLOCKED.
Triple X time	$X < W$	Some detailed rule $X < 1/2 W$	Context-specific conflict.
Z failure	31.8 in one source	Nearby 38.2 note for related case	TBD.
LD/ED same impulse	Ambiguous wording	Other examples may imply different behavior	Owner decision.
Timeframe degree mapping	Daily/75m/15m	Holding-period dependent mappings	Degree Mapping Preset.
Log chart	Not universal in Chartking spec	Hardik says always log	Optional source preset.
MACD parameters	Unspecified	Examples often show 12/26/9 visually	Inputs; never claim source mandated 12/26/9.
Pivot algorithm	No exact source algorithm	Visual/manual notes	Owner-approved raw pivot algorithm; expose sensitivity.

Y. Recommended Pine v6 Module Architecture

The developer is a Pine Script expert, so the following is architecture-level pseudocode, not intended as copy/paste syntax. The goal is to keep the theory isolated in testable modules.

```

MODULE 01 - RawPivotEngine
    detect pivots, ATR/range filter, confirmed/raw buffers

MODULE 02 - DegreeRouter
    map chart/source timeframe, Important H/L, degree metadata

MODULE 03 - BaseEngine
    base candidates, MACD/divergence evidence, base lock

MODULE 04 - ImpulseEngine
    w1/w2/w3/w4/w5 candidates, normal/terminal subtype

MODULE 05 - ExtensionEngine
    internal move counts 5/9/13/17/21, degree fall

MODULE 06 - DiagonalEngine
    LD / ED

MODULE 07 - CorrectionEngine
    ZigZag / Flat / ABC / WXY / Triple / Triangle

MODULE 08 - FibEngine
    anchor-specific levels + ratios

MODULE 09 - TimeEngine
    source presets, expected windows, Fib time zones

MODULE 10 - SupportEngine
    MACD/RSI divergence, gaps, candles, trendlines/channels

MODULE 11 - RecountEngine
    locked confirmed count, alternates, hard invalidation, FBD/FB0 candidate origin

MODULE 12 - DisplayDebug
    labels, source rule IDs, next condition, pass/fail numbers

```

Y1. Suggested Candidate Types

```

enum Status { SEARCHING, FORMING, CONFIRMED, ALTERNATE, INVALID, TBD_BLOCKED }

type Pivot
    int bar
    float price
    int side
    int degree
    bool confirmed
    float macdLine
    float macdHist
    float rsi

type WaveCandidate
    string id
    string label
    string subtype
    int degree
    int direction
    Pivot start
    Pivot finish
    Status status
    float fibRetrace
    float fibExtension
    int duration
    int internalCount
    string invalidReason
    string sourceRuleId

```

Y2. Functions to Add / Refactor

Function	Purpose
f_eval_base()	Return base candidate state and reasons.
f_eval_w1()	Build/validate W1 and degree confirmation.
f_eval_w2()	Evaluate W2 price/time/internal correction and origin invalidation.
f_eval_w3()	Trending/terminal/extension classification.
f_eval_w4()	Open/track correction child candidates and subtype-aware overlap.
f_eval_w5()	Normal/truncated/extended/ED logic plus divergence.
f_eval_zigzag() / f_eval_flat()	Simple correction classifiers.
f_eval_double() / f_eval_triple()	Complex correction containers.
f_eval_triangle_*	Six triangle subtype classifiers.
f_eval_ld() / f_eval_ed()	Diagonal parsers.
f_apply_hard_invalidations()	Central hard-rule rejection with reason codes.
f_rank_candidates()	Choose primary/alternate without deleting still-valid correction candidates.
f_controlled_recount()	Promote alternate/new base only on approved trigger.
f_debug_rule()	Emit rule ID, numeric inputs and pass/fail.
f_draw_candidate()	Draw based on FORMING/CONFIRMED status, not fixed sequence.

Y3. Main Loop Pseudocode

```

on each closed bar:
    update indicators and source timeframe contexts
    rawPivotEvent = RawPivotEngine.update()

    if rawPivotEvent:
        DegreeRouter.update(rawPivotEvent)
        BaseEngine.update(rawPivotEvent)

        for each active degree:
            ImpulseEngine.update(rawPivotEvent, degree)
            CorrectionEngine.update(rawPivotEvent, degree)
            ExtensionEngine.update(rawPivotEvent, degree)
            DiagonalEngine.update(rawPivotEvent, degree)

        HardRuleEngine.invalidateImpossibleCandidates()
        CandidateRanker.updatePrimaryAndAlternates()
        RecountEngine.applyOnlyIfTriggerApproved()

    DisplayDebug.renderFromState()

```

Z. Step-by-Step Refactor Plan for the Existing Script

Sprint	Developer task	Do not move forward until
Sprint 0 - Freeze baseline	Save current script/Python outputs and 10 known charts as regression fixtures.	Baseline screenshots and outputs are archived.
Sprint 1 - Decouple labels	Keep raw pivots but stop modulo phase from deciding Elliott labels. Add candidate/status structs.	No main label appears merely because pivot index advanced.
Sprint 2 - Base engine	Implement degree-aware Important H/L, oscillator evidence, locked base and alternate origin.	Base remains stable on minor swings; 144-day test passes.
Sprint 3 - W1/W2	Implement W1 degree + W2 correction container, source-mode W2 range/time and hard origin invalidation.	Normal/microscopic/deep/invalid W2 fixtures pass.
Sprint 4 - W3	Implement Trending/Terminal/Extension + internal counts + MACD support.	Small pivots inside extended W3 no longer become W4/W5.
Sprint 5 - W4/W5	Correction container, overlap subtype, W5 normal/truncated/ED.	Premature W5 bug is gone without forced A-B-C sequencing.
Sprint 6 - Corrections	ZigZag, Flat, WXY, Triple, triangles.	Manual correction dropdown no longer controls classification.
Sprint 7 - Time/support	Time presets, trendlines/channels, gaps, MACD/RSI support.	Debug shows source rules and no support module overrides hard price structure.
Sprint 8 - Project overlays	HP BUY/SELL, FBD/FBO controlled recount, C/Z completion event.	Signals consume Elliott state but do not mutate the count.
Sprint 9 - Python parity	Port same state model to Python and replay fixtures.	Pine/Python confirmed pivots materially match; differences explained.
Sprint 10 - Production QA	Performance, object limits, dashboard, non-repaint proof.	Full acceptance suite passes.

Z1. Definition of “Premature Wave 5 Bug Fixed”

Acceptance rule

The bug is only considered fixed if a complex W4 example can add several internal pivots (A/B/C, W/X/Y, or triangle legs) without the parent count becoming Wave 5 until the W4 correction itself is confirmed complete. Simply inserting a fixed A-B-C phase before W5 is not sufficient.

Z2. Definition of “B Wave up to 111% Fixed”

Acceptance rule

The fix is valid only if 61.8%-111% is applied inside the Flat candidate, while a Zig-Zag B uses its own source range and complex corrections remain possible. Above the Flat maximum must invalidate/reclassify the Flat candidate instead of leaving a stale B label.

Z3. Definition of “Non-Repainting”

Acceptance rule

Raw/current FORMING legs may update until confirmed. Once a main-degree label is CONFIRMED, it must not move because of a small later pivot. A confirmed label may only change through a logged controlled recount caused by a hard structural invalidation or approved higher-degree/origin event.

AA. Detailed Developer Debug Panel Specification

Row	Example value	Why it matters
DEGREE / TF	Intermediate / 75m	Prevents degree mixing.
BASE	LOCKED @ 2385.20 / bar 1234	Shows origin stability.
CURRENT PARENT	W4 FORMING	Parent context.
PRIMARY PATTERN	Flat FORMING	Main correction candidate.
ALTERNATE	Double WXY FORMING	Ambiguity kept visible.
RAW PIVOTS	27 / main pivots 4	Shows raw vs semantic separation.
FIB ANCHOR	0-3 = 2385.2 -> 2455.8	Makes formulas auditable.
FIB VALUE	W4 retrace 34.6% PASS	Numeric source check.
TIME RULE	Charting W4: nearest 2xW2, error 7%	Shows selected time source.
MACD	W4 low support PASS	Supporting evidence.
HARD RULE	Normal overlap: N/A - terminal candidate active	Prevents wrong invalidation.
NEXT CONDITION	Need C internal completion + 0-B break	Developer/user knows what is missing.
LAST INVALID	ZigZag B range failed @ 69.2%	Explains why candidate was removed.
RECOUNT	NONE / last reason	Audit non-repaint behavior.

AB. Extended Acceptance Tests for a Developer Who Does Not Know Elliott

ID	Scenario	Input	Expected
E01	Raw pivot vs Elliott pivot	Insert 8 small pivots inside a strong W3	Raw array grows; main degree still W3.
E02	Base replacement	New important low appears while count remains valid	Open alternate base; do not replace locked base.
E03	Base hard invalid	Price invalidates current origin	Log reason; promote valid alternate/recount.
E04	W1 subordinate	Five moves complete before degree threshold	Mark subordinate/FORMING; do not confirm degree W1.
E05	W2 microscopic modes	14%-23.6% retracement under three source presets	Correct status differs visibly by preset.
E06	W2 time conflict	Actual W2 = W1	Different source presets show pass/fail/TBD rather than hidden decision.
E07	W3 terminal	W3 five moves at 100% extension	Terminal candidate remains; no forced trending W3.
E08	W3 extension	13 internal moves	Parent label remains W3 until degree-fall/completion.

ID	Scenario	Input	Expected
E09	W4 overlap normal	W4 enters W1 territory in normal impulse	Normal candidate invalid.
E10	W4 overlap diagonal	Same price path but valid diagonal context	Diagonal candidate remains valid.
E11	W4 Double	WXY with 5 raw turning points	Parent stays W4 until Y completion.
E12	W4 Triangle	ABCDE legs	No W5 until E/triangle completion.
E13	Flat vs ZigZag B	B=45%, then another sample B=90%	First favors ZigZag; second Flat; no universal B rule.
E14	Flat B >111	B=118%	Flat invalid/reclassified; stale B removed.
E15	C Fib hit without internals	C touches target early	Keep C FORMING if internal completion not satisfied.
E16	Double confirmation TBD	Valid WXY but source ??? not resolved	Show TBD-BLOCKED / source conflict.
E17	Triple XX	XX violates X boundary	Reject Triple candidate with exact reason.
E18	Triangle location	ABCDE appears in W2	Triangle candidate rejected because source excludes W2.
E19	ED position	3-3-3-3-3 appears in W3	Do not label ED if source position invalid.
E20	W5 divergence	Price makes higher W5 high but MACD lower than W3	Divergence support PASS.
E21	W5 truncation	Double-extension fixture, W5 below127	Truncated subtype possible; normal W5 candidate may fail without killing impulse.
E22	Time origin Flat	Flat base differs from market high used for time	Price/time origins stored separately.
E23	144-day consistency	Same symbol on D/1h/15m	Important H/L source context consistent.
E24	Pine/Python	Replay fixture	Same confirmed pivot IDs and pattern states.
E25	Recount audit	Force hard invalid after several confirmed labels	Log old/new base, reason, affected labels; no silent rewrite.

AC. What the Developer Must Not Do

- [] NEVER: Keep f_phase/modulo as the real wave engine and only add more warning conditions.
- [] NEVER: Treat every accepted ZigZag swing as a new main-degree Elliott wave.
- [] NEVER: Use 61.8%-111% as a universal B-wave rule.
- [] NEVER: Use 14% as a universal W2/W4 normal minimum without a source preset/subtype.
- [] NEVER: Hard-code one time rule from conflicting sources and hide the decision.
- [] NEVER: Confirm Wave C because the third corrective pivot appeared.
- [] NEVER: Reject all Wave-4 overlap before checking terminal/diagonal subtype.
- [] NEVER: Call a wedge a Leading Diagonal without its position/internal/Fib rules.
- [] NEVER: Call 3-3-3-3-3 an Ending Diagonal outside W5/C.
- [] NEVER: Delete alternate correction candidates just because one pattern currently scores higher.
- [] NEVER: Reset the base on every new Important H/L or FBD/FBO event.
- [] NEVER: Let trade signals mutate the Elliott structure.

- [] NEVER: Claim “non-repaint” if confirmed labels silently move with later raw pivots.
- [] NEVER: Let Pine and Python implement different pivot/state rules and then compare only screenshots.

AD. Final Instruction to the Developer

Implementation goal

You do not need to become an Elliott Wave trader. You need to implement a deterministic state machine that preserves the supplied rules exactly, exposes conflicts instead of guessing, and provides enough debug information that the owner can compare the automatic count with manual Elliott marking.

Order of authority

1) Hard price/position/internal-structure rule. 2) Degree and parent context. 3) Pattern-specific Fib. 4) Time rule under selected source preset. 5) MACD/RSI/trendline/gap/candle support. 6) Trade overlay. A lower item must not override a hard failure in a higher item.

Completion standard

Do not declare the feature complete when labels look visually plausible. Declare it complete only when every confirmed label has a traceable candidate history, Fib/time anchors, source rule IDs, invalidation conditions, and matching Pine/Python result on the acceptance fixtures.

AE. Source Visual Index Used in This Addendum

Visual	Source
Wave 1 degree/base visual	EW Notes - Chartking v2, p.6
Wave 2 Fib/time visual	EW Notes - Chartking v2, p.8
Wave 3 extension/time visual	EW Notes - Chartking v2, p.11
Wave 4 alternation visual	EW Notes - Chartking v2, p.13
Wave 5 projection/divergence visual	EW Notes - Chartking v2, p.15
Leading Diagonal visual	EW Notes - Chartking v2, p.18
Ending Diagonal visual	EW Notes - Chartking v2, p.22
Zig-Zag normalized table	Developer Requirement Specification, p.14
Flat normalized table	Developer Requirement Specification, pp.15-16
Double WXY table	Developer Requirement Specification, p.19
Triple WXYXXZ table	Developer Requirement Specification, p.22
Triangle family summary	Developer Requirement Specification, p.24
Trendline/channel summary	Developer Requirement Specification, p.32
Candidate state engine	Developer Requirement Specification, p.39
Time-analysis source visuals	Time Analysis, pp.1-4
Hardik W1 time / trendline visuals	ELLIOT WAVE-HARDIK PAVAGADHI, pp.14,39

END OF DETAILED IMPLEMENTATION ADDENDUM

17. Recommended indicator state engine (implementation translation)

This section is recommended software architecture, not a new trading rule. It translates the source into a deterministic implementation that can handle incomplete patterns without repainting historical labels arbitrarily.

17.1 Candidate lifecycle

State	Meaning
SEARCHING	No valid base/current candidate yet.
FORMING	Enough structure exists to propose a wave/pattern, but all confirmation conditions are not met.
CONFIRMED	Source-defined structural, fib/time and required confirmation conditions are met.
ALTERNATE	A secondary count remains possible because correction rules are guideline based.
INVALID	A hard source rule has been violated. Store the exact reason.
TBD-BLOCKED	A decision depends on a source blank/conflict; show it rather than guessing.

17.2 Evaluation order

- **Step 1:** determine analysis timeframe / degree and locate Important High-Low plus candidate base point.
- **Step 2:** create impulse candidate and subordinate pivots; apply hard price-structure rules before supporting indicators.
- **Step 3:** calculate Fib retracement/extension and time relationships at every confirmed pivot.
- **Step 4:** classify Trending vs Terminal impulse, diagonal subtype and extension subtype if applicable.
- **Step 5:** when correction begins, maintain Zig-Zag / Flat / Double / Triple / Triangle candidates in parallel.
- **Step 6:** use MACD, gap, candlestick, channel and trendline events as the source describes - primarily as confirmation/ranking, not as replacements for wave structure.
- **Step 7:** confirm/invalidate labels only after the relevant source rule is satisfied/violated; retain reason codes for debugging.

17.3 Suggested internal record

```
WaveCandidate { degree, timeframe, direction, label, patternType, startBar, endBar,
startPrice, endPrice, durationBars, fibRetrace, fibExtension, internalCount, macdState,
divergenceState, gapState, candlestickState, trendlineState, channelState, status,
invalidReason, sourceRuleId }
```

18. Chart display requirements (recommended handoff)

Element	Required display behavior
Wave labels	Show 0/Base,1,2,3,4,5,A,B,C,W,X,Y,XX,Z and A-B-C-D-E triangle labels at confirmed pivots; optional dotted label for forming candidates.
Pattern name	Display active structure: Impulse, Terminal, Leading Diagonal, Ending Diagonal, Zig-Zag, Flat subtype, Double, Triple, Triangle subtype, Extension subtype.
Fib levels	Draw only the levels relevant to the current candidate and source rule; label the anchor pair (e.g..0-1-2,0-A-B,0-W-X,0-Y-XX).
Trendlines/channels	Draw source 0-2,1-3,2-4,0-B and triangle boundaries with anchor labels and break state.
MACD support	Display compact status (lowest/highest/top/bottom, +div/-div) rather than replacing wave labels.
Gap / candle support	Small labels at the source-defined context.
Status panel	Show current degree, current wave, candidate pattern, confirmed/forming/alternate/invalid, next source condition and conflict/TBD if applicable.
Debug mode	Developer-only mode should print rule IDs passed/failed and numeric ratios for backtesting/validation.

19. Inputs / settings required for implementation